

International Effects of Corporate Tax Cuts on Income Distribution^{*}

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Abstract

This paper evaluates the domestic and cross-country effects of corporate income tax (CIT) cuts on income inequality using a tractable multi-country DSGE model calibrated to U.S. and Canadian data. We illustrate that the cross-border propagation of differentiated equity holdings can help explain the increase in income inequality, both within- and cross-country, in response to a CIT cut. A lower barrier to capital mobility alleviates this divergence in income, while lower trade costs exert little effects on income distribution. The model is also consistent with recent evidence that middle-income workers experience lower income growth than their counterparts in either extreme of the skill spectrum.

Keywords: Cross-country investment, income inequality, corporate tax cuts, the Tax Cuts and Jobs Act of 2017.

JEL Classification Numbers: F38, F23, D63.

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1 Introduction

In late December 2017, the United States slashed its corporate income tax rate (CIT) and became the latest example of a downward trend across OECD countries since the early 1980s (see Figure 1).¹ The passage of this CIT cut has rekindled the debate over who stands to benefit from it and whether such a type of tax cuts exacerbates income inequality.² While a large body of literature has focused on the burden of CIT on factors of production (i.e., capital and labor), studies that assess the cross-border impacts of national CIT cuts on income inequality have been few and far between.³

This paper contributes to such a debate by presenting a multi-country DSGE model with heterogeneous agents to explore the effects of a unilateral CIT cut on domestic income inequality and, the often overlooked, international spillover effects. In particular, the model provides a mechanism by which a CIT cut increases income inequality, domestically and internationally, and examines how such increases are driven over the short and long runs. Our model consists of “capitalists” households, who own capital, and “worker” households whose income consists exclusively of labor income. Our model shows that capitalists, over the short run, capture *more* than the full amount of the benefits that a CIT cut provides to the country’s output. In the long run, our model also predicts increased inequality as capitalists around the world and the workers of the country with the CIT cut enjoy higher income.

A CIT cut can increase inequality in two distinct manners and, absent any normative judgment of inequality per se, one of the two is welfare-diminishing for the households relying solely on labor income. The first way in which inequality could increase is if the aggregate benefits from a CIT cut (for example, via higher output and investment) are disproportionately captured by the “capitalist” households, while the “worker” households also benefit from a higher wage. Our model predicts this type of increase in the long run. The second way in which inequality could increase is when “capitalists” can capture the entirety of the aggregate benefits from the tax cut while “workers” experience a reduction, or no change, in their income due to a wage decrease.

¹The Tax Cuts and Jobs Act of 2017. H.R.1 — 115th Congress (2017-2018). Enacted on 12/20/2017. The act reduced the corporate tax rate from a maximum of 35% to a 21% rate for tax years beginning after 2017.

²See, for example, [Piketty and Saez \(2003\)](#) or more recently [Smith et al. \(2019\)](#) and [Altig et al. \(2019\)](#).

³Recent empirical studies explore the relationship between corporate taxes and inequality (e.g., [Suarez Serrato and Zidar \(2016\)](#); [Nallareddy et al. \(2018\)](#)), with a common focus on *within*-country analyses using local CITs.

The short-run dynamics of our model predict this welfare-diminishing type of income inequality increase following the CIT cut.

Our multi-country DSGE model features two types of households (workers and capitalists), imperfect competition, cross-border investment, and incomplete markets. Workers' only source of income comes from working at the domestic firm while capitalists' income consists of labor income, capital rental income, and after-tax profit distributions from the firms to which they supplied capital. Each country hosts a representative firm that sells its product in the domestic and foreign markets under monopolistic competition. Firms' profits are subject to corporate income taxes in their country of residence. In every period, the *non-zero* after-tax profits are distributed among the investors based on the ratio of the capital they provided to the total capital input used by the firm. Capitalists can invest both domestically and internationally, but foreign investors face a higher adjustment cost than domestic investors.⁴ We illustrate the implications of the model using a two-country version calibrated to U.S. and Canadian data.

We depart from previous models in the corporate tax incidence literature along several dimensions. First, we do not impose a small open economy assumption, which allows the rental rates in our model to respond to tax changes occurring not only in the source country but also in its trading partners. This transmission channel amplifies the spillover effects of capital reallocation. Second, we depart from perfect competition and assume monopolistic competition, which makes firms' profits a non-zero component of the capitalists' income.⁵ In other words, the composition of the capitalist's income goes beyond labor and capital rental income to include a business/equity income component. This component is critical in understanding the changing landscape of the top-income composition in the United States (see, e.g., [Smith et al. \(2019\)](#)) and in explaining the increases in income inequality observed after a corporate tax cut (see, e.g., [Nallareddy et al. \(2018\)](#)). Finally, the static nature of some earlier models provided no insight into the dynamic responses following an unexpected tax change or an aggregate shock. This shortcoming, as carefully documented by [Auerbach \(2006\)](#), is addressed by the use of a dynamic setting in our paper.

⁴See [Kollmann et al. \(2014\)](#), or [Hall \(2011\)](#) for evidence on incomplete markets and cross-country investment flows.

⁵Previous models departing from the perfectly competitive markets include [Davidson and Martin \(1991\)](#); [Liu and Altshuler \(2013\)](#), who use a (duo)oligopoly model with a repeated game Nash equilibrium; and [Suarez Serrato and Zidar \(2016\)](#) who also assume a monopolistic competition structure.

Our model dynamics predict that a unilateral and unexpected CIT cut results in higher income inequality, in all countries, and an initial income decrease for the worker living in the country with the tax cut. Following the CIT cut, income for capitalists in all countries increases significantly. Moreover, these increases are driven by the gains in equity income, which in turn reduce the capitalists' labor supply. In stark contrast, the worker in the country with the tax cut experiences an income decrease as their real wage falls upon impact and slowly reverts to its ex-ante level.

The mechanism driving the income and inequality dynamics above is the reallocation of resources to supply more capital to the firm in the country with the newly reduced CIT rate. Intuitively, a reduction in the CIT rate raises the firm's after-tax profit, which makes investing in the firm more attractive. Capitalists, domestic and foreign alike, react to this new environment by ramping up their investments, which drives down rental rates. Since the firm is subject to lower capital costs, it substitutes labor with capital, which leads to the initial wage decrease. While both the worker and the capitalist suffer a wage decrease, the increase in the capitalist's equity income more than offsets their losses from labor income and thus, the capitalist's wage recuperates at a substantially faster pace than the worker's. The overall effect is an increase in income inequality. The CIT cut also has spillover effects in other countries as it boosts the income of their capitalists but has almost no effect on the workers' income. Consequently, income inequality increases in other countries as well.

In the long run, a CIT cut also increases income inequality across countries but, unlike in the short run, the domestic worker's income experiences a small increase while the foreign counterparts experience a minor decrease. In particular, the CIT cut induces the reallocation of capital into the country that enacts the tax cut, thereby increasing its output at the expense of its trading partners' output. Even though output increases for only one country, the net income of capitalists across the world increases, which implies that equity income gains dominate labor income losses that arise from the output decrease. In contrast, only the worker in the country with the CIT cut experiences an increase in their income, while workers in other countries experience a minor decrease in their income.

We examine the sensitivity of our results under the benchmark model in several directions. First, we transfer the entire corporate tax revenue to the worker. The model predicts responses

more pronounced than those of the benchmark: the negative impact on the worker of the country with the tax cut is amplified and takes much longer to recover than in the benchmark model. This result is not surprising since, under this specification, the CIT cut has a direct effect on the worker’s income through the reduction in transfers.

Next, we consider the role of globalization by comparing the benchmark model with two specifications that represent isolationist policies: (i) higher trade costs and, (ii) higher cross-border capital adjustment costs for foreign investors. Overall, we find lower barriers to capital mobility dampen the increase in income inequality in the country with the CIT cut while having a mixed effect on the trading partner. Increased trade costs impact the two types of agents equally and thus exert little effects on income distribution.

Finally, we augment our benchmark model to include a “service occupations” sector whose output relies solely on manual labor from a new agent: the service worker household. Our motivation for this extension follows from [Autor and Dorn \(2013\)](#), who document a significant polarization of labor and wages in the United States using data from 1980 to 2005. Overall, we find that our model, when augmented with these non-corporate services workers, is consistent with the “hollowing out” effect documented by [Autor and Dorn \(2013\)](#) in which workers in the middle of the skill spectrum experienced lower income growth than their counterparts that belong to either extreme of the skill spectrum.⁶ Intuitively, in the country with the CIT cut, two opposing mechanisms drive the demand for services. First, the capitalists’ increase in income leads to an increase in their demand for services. Second, there is a decrease in demand from the traditional/non-service worker households. As a result, while the wage of the service workers (i.e., the low-skill workers) decreases on impact, the magnitude of such a response is between those of the worker households (i.e., the middle-skill workers) and those of the capitalists (i.e., the high-skill workers).

Our paper relates to the longstanding literature on corporate tax incidence. This literature starts with the seminal work by [Harberger \(1962\)](#), who uses a closed-economy general equilibrium framework to show that capital bears the entire burden of corporate taxes. Subsequent papers have taken the general equilibrium approach into an open-economy framework (see, e.g., such as [Mutti and Grubert \(1985\)](#), [Kotlikoff and Summers \(1987\)](#), [Gravelle and Smetters \(2006\)](#) or [Harberger](#)

⁶We are especially grateful to an anonymous referee and the editor for suggesting the addition of such a type of worker.

(2008)) to find that capital and labor, as factors of production, both share the burden of corporate taxes.⁷ While also examining the interplay between capital and labor as factors of production in response to a CIT cut, our paper complements this literature by allowing for heterogeneous agents and by focusing on both the domestic and cross-country responses of income inequality. Another source of novelty in our paper is to highlight the importance of cross-border transmission of (non-zero) dividends in understanding these inequality responses. Our model’s implications for inequality and income dynamics both support and complement recent studies (see, for example, [Suarez Serrato and Zidar \(2016\)](#), [Nallareddy et al. \(2018\)](#), and [Fuest et al. \(2018\)](#)).⁸

The remainder of the paper is as follows. Section 2 describes an N –country version of the model. Section 3 discusses our calibration of a two-country version of the model to U.S. and Canadian data. Section 4 examines the properties of the model to an unexpected corporate tax cut and Section 5 provides some extensions to the baseline model. Section 6 analyzes the long-run effects of a unilateral corporate tax cut. Section 7 concludes.

2 Model

There are N countries, and within each one resides a representative firm and two types of households: a worker and a capitalist. Both households supply labor to the domestic firm and have the same functional form for preferences, but only capitalists can supply capital to firms in any country.⁹ In return for their investments, the capitalists receive a share of after-tax profits from the firms in which they invested. This distribution is proportional to their shares of capital holding.

We use pairs of subscripts (ij) to keep track of the origin and the destination country for

⁷[Auerbach \(2018\)](#) provides a thorough and critical review of the corporate tax incidence literature, as well as an insightful discussion regarding the effects of the corporate tax changes of the Tax Cuts and Jobs Act of 2017.

⁸[Nallareddy et al. \(2018\)](#) evaluate the short-to-medium term impact of states’ CIT cuts and find evidence that such cuts raise income inequality. They find only households with income above \$200,000 saw an increase in their income. The increase was driven by higher equity income and capital returns, which offset the decline in their labor income. [Fuest et al. \(2018\)](#) also analyze the short-term effects of local CIT changes using data from 3,522 German municipalities. They find that a significant portion of the tax burden is passed on to workers, thus conflicting with the short-term dynamics of our model. [Suarez Serrato and Zidar \(2016\)](#) consider the effect local corporate taxes from a longer time perspective (10-year period) and find that business owners capture the majority of the benefit, but workers also benefit from higher wages.

⁹The assumption that one of the two households, or half of the population in each country, invests in firms simplifies the solutions and expositions of the results. The data lend some credibility to this assumption since about half of the U.S. population participated in the stock market, according to the 2016 Survey of Consumer Finances. Participation included people with direct holdings and also those with indirect holdings (such as mutual funds and retirement plans).

variables that flow across countries. In what follows, a variable $x_{ij,t}$ conveys that x is produced in (comes from) country j and is consumed (used) in country i . We use single subscripts to denote national variables; e.g., $Y_{j,t}$ is country j 's output at time t .

To economize on space, we only describe the model from the perspective of the agents residing in country j . The problem and solution for agents in other countries are analogous.

2.1 Households

Workers

Workers maximize expected lifetime utility of the form:

$$\max \mathbb{E} \sum_{t=1}^{\infty} \beta^t \left[\frac{(C_{j,t}^w)^{1-\psi}}{1-\psi} - \chi \frac{(L_{j,t}^w)^{1+\xi}}{1+\xi} \right], \quad (1)$$

where β is the subjective discount factor, $L_{j,t}^w$ is labor supplied by the workers, ψ is the degree of risk aversion, ξ is the inverse of the Frisch elasticity, and $C_{j,t}^w$ is a composite consumption good defined as:

$$C_{j,t}^w = \left[\sum_{i=1}^N (c_{ji,t}^w)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad (2)$$

where $c_{ji,t}^w$ denotes consumption of the goods produced in country i by the worker in country j . Equation (2) is an Armington aggregator with an elasticity of substitution σ and no home bias.

The budget constraint for the workers is simply

$$\sum_{i=1}^N p_{ji,t} c_{ji,t}^w = \omega_{j,t}^w L_{j,t}^w, \quad (3)$$

where $p_{ji,t}$ is the price paid by consumers in j for the goods produced in i , and $\omega_{j,t}^w$ is the wage rate paid by the domestic firm to the worker type household in country j . The lack of investment

or saving opportunities implies that workers make intra-temporal decisions only.¹⁰

Per [Dixit and Stiglitz \(1977\)](#), the demand for each good is expressed in terms of the composite good ($C_{j,t}^w$) and the price index ($P_{j,t}$):

$$c_{ji,t}^w = \left(\frac{p_{ji,t}}{P_{j,t}} \right)^{-\sigma} C_{j,t}^w, \quad \forall i = 1, \dots, N. \quad (4)$$

$$P_{j,t} = \left[\sum_{i=1}^N p_{ji,t}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \quad (5)$$

Capitalist

Capitalist's preference is identical to that of the worker:

$$\max \mathbb{E} \sum_{t=1}^{\infty} \beta^t \left[\frac{(C_{j,t}^c)^{1-\psi}}{1-\psi} - \chi \frac{(L_{j,t}^c)^{1+\xi}}{1+\xi} \right], \quad (6)$$

where $C_{j,t}^c$ is a composite consumption good aggregated in a similar fashion as in Equation (2), $L_{j,t}^c$ is the amount of labor hours supplied by the capitalist household to the domestic firm. The budget constraint faced by the capitalist is:

$$\sum_{i=1}^N p_{ji,t} c_{ji,t}^c + \sum_{i=1}^N p_{ii,t} I_{ij,t} = \omega_{j,t}^c L_{j,t}^c + \sum_{i=1}^N r_{i,t} k_{ij,t} + \sum_{i=1}^N s_{ij,t} (1 - \tau_{i,t}) \pi_{i,t}. \quad (7)$$

The right-hand side of Equation (7) introduces the various income sources of the capitalist household. First, the capitalist supplies labor to the domestic firm at a wage rate $\omega_{j,t}^c$, which may differ from the rate received by the worker households ($\omega_{j,t}^w$). Capitalists supply capital to firms and receive factor payments ($r_{i,t}$), per unit of capital, in the next period. Rental rates are country-specific since consumption-investment goods are not homogeneous across countries. The remaining source of income is a fraction of the after-tax profits from firms in which the capitalist

¹⁰We explored the inclusion of risk-free bonds available to workers and capitalists that cleared at the national or world level. As shown by [Schmitt-Grohe and Uribe \(2003\)](#), additional restrictions must be imposed to avoid the unit root that arises when bond interest rates are the same across different agents. The cleanest solution to this problem is to include a premium on the risk-free interest that is based on the agent's level of debt; however, this approach only works in the context of small open economies. In the case of our model, we would need to include another agent in the economy, e.g., a bank that profits from the interest rate differential between borrowers and lenders. This added layer of complexity, while interesting, is not of first-order importance for the question at hand, namely, how corporate tax cuts affect households that own assets around the world versus those that do not.

invested. We assume that firms' profits ($\pi_{i,t}$) are subject to corporate income taxes, at the rate of $\tau_{i,t}$, in the country where the firm produces.¹¹ Also, we do not allow firms to retain and invest their earnings; instead, we assume that all earnings are distributed to the owners based on the proportion of capital they provide. The share of after-tax profits corresponding to each capitalist household ($s_{ij,t}$) is simply the ratio of capital they provided ($k_{ij,t}$) over the firm's total capital ($K_{i,t}$):

$$s_{ij,t} = \frac{k_{ij,t}}{K_{i,t}} \quad \text{for } i = 1, \dots, N \quad (8)$$

$$K_{i,t} = \sum_{j=1}^N k_{ij,t} \quad \text{for } i = 1, \dots, N \quad (9)$$

Capitalists spend their income on consumption, on investments in firms, and on the payment of the capital adjustment costs. The capital input used by the firm in country i cannot be substituted with goods produced in other countries. We assume that purchases of goods for investment purposes occur in the destination country, thus avoiding transportation costs. Therefore, a capitalist in country j pays $p_{ii,t}$ per unit of net investments made to the firm in country i in period t ($I_{ij,t}$). We included a quadratic adjustment cost on net investments, which implies the following capital evolution equation:

$$I_{ij,t} = k_{ij,t+1} - (1 - \delta)k_{ij,t} + \Phi(\eta_{ij}, k_{ij,t}), \quad (10)$$

$$\Phi(\eta_{ij}, k_{ij,t}) = \frac{\eta_{ij}}{2} (k_{ij,t+1} - k_{ij,t})^2, \quad (11)$$

where $\eta_{ij} > \eta_{ii}$ for all $i \neq j$, i.e., foreign investors face costlier adjustment, per unit of net investment, than domestic investors. Examples of extra costs faced by foreign investors include monitoring the investment, executing the transaction, research costs, and the risk premia associated with making cross-border investments.

¹¹In tax parlance, we impose a worldwide tax system on firms' profits. The United States followed this system until the Tax Cuts and Jobs Act, after which the United States adopted a quasi-territorial tax system. Under the territorial system, only profits generated within a country's borders are subject to the country's corporate tax rate. Changing from a worldwide taxation system to a territorial system impacts the incentives of firms to repatriate profits. This is an interesting yet complex issue that is also important for our understanding of international corporate tax competition. We leave this issue for future research.

2.2 Firms

Each country hosts a representative firm that produces under the technology:

$$Y_{j,t} = A_{j,t}(K_{j,t})^\alpha (L_{j,t}^c)^\kappa (L_{j,t}^w)^{1-\alpha-\kappa}, \quad (12)$$

in which capital input ($K_{j,t}$) can be provided by domestic and foreign capitalists; $L_{j,t}^c$ is labor provided by the domestic capitalist, and $L_{j,t}^w$ is labor provided by the domestic worker.

Since firms sell their products in monopolistically competitive markets, they set prices to maximize profits. The optimal prices charged in the domestic market ($p_{jj,t}$) and for exporting to country i ($p_{ij,t}$) satisfy:

$$p_{jj,t} = \frac{\sigma}{\sigma - 1} \Omega_{j,t} \quad (13)$$

$$p_{ij,t} = \theta_{ij} p_{jj,t} \quad \forall i \neq j, \quad (14)$$

where $\theta_{ij} > 1$ is an iceberg trade cost incurred by exporting goods from country j to i , and $\Omega_{j,t}$ is the marginal cost equal to the Lagrange multiplier associated with the cost minimization problem:¹²

$$\begin{aligned} \min_{K_{j,t}, L_{j,t}^c, L_{j,t}^w} \quad & r_{j,t} K_{j,t} + \omega_{j,t}^w L_{j,t}^w + \omega_{j,t}^c L_{j,t}^c, \\ \text{s.t} \quad & Y_{j,t} \geq Y_j. \end{aligned} \quad (15)$$

2.3 Government

The government collects corporate income taxes and spends them on pure government consumption ($G_{j,t}$) that does not affect the households' utility.¹³ We allow governments to sell (buy) bonds ($B_{j,t}^g$)

¹²While multiple papers using monopolistic competition (e.g., [Krugman \(1979\)](#) or [Melitz \(2003\)](#)) include fixed and/or entry costs, we abstract from including those in this paper. Entry costs would be necessary if our model considered firm entry and/or exit. Having this feature would require a global solution method that severely restricts the number of state variables. The inclusion of fixed costs would provide a lower bound for profits but would not change the model dynamics except when the equilibrium solution is sufficiently close to the corner case (i.e., steady-state profits equal fixed costs). It is also worth noting that most of the papers that include fixed and/or entry costs only focus on steady-state analysis with the notable exception of [Ghironi and Melitz \(2005\)](#).

¹³We later relax this assumption and allow the workers to receive tax revenue (i.e., $\tau_{j,t}\pi_{j,t}$) as a transfer.

to finance the windfall (excess) between tax revenue and tax expenditure. We assume that the cost of borrowing increases as the country's debt becomes larger. In particular, the government's budget constraint is:

$$p_{jj,t}G_{j,t} - \tau_{j,t}\pi_{j,t} + B_{j,t}^g r_{j,t}^b = B_{j,t+1}^g - B_{j,t}^g, \quad (16)$$

where

$$r_{j,t}^b = r^{rf} + \psi_2(e^{B_{j,t}^g - \bar{B}_j} - 1), \quad (17)$$

in which $r^{rf} = \frac{1}{\beta} - 1$ is the risk-free interest rate, ψ_2 dictates how costly it is for the country to deviate from its long-run debt level ($\bar{B}_j$). We posit government spending as a mean-reverting AR(1) process as follows¹⁴

$$\log(G_{j,t}) = (1 - \rho_G) \log(\bar{G}_j) + \rho_G \log(G_{j,t-1}) + \varepsilon_{j,t}^G. \quad (18)$$

We model corporate tax rates as a mean-reverting stochastic process of the following functional form

$$\log(\tau_{j,t}) = (1 - \rho_\tau) \log(\bar{\tau}_j) + \rho_\tau \log(\tau_{j,t-1}) + \varepsilon_{j,t}^\tau, \quad (19)$$

in which $\bar{\tau}_j$ denotes the steady-state tax rate in country j and ρ_τ denotes the persistence of the corporate tax rate process. Here $\varepsilon_{j,t}^\tau$ is i.i.d. and follows $N(0, \sigma_j^\tau)$ in which σ_j^τ is finite.

2.4 Market Clearing

In addition to the standard labor and capital market-clearing conditions, the balance of payments (BoP) equation must hold. The BoP is the zero-sum of the current account (CA) and the capital

¹⁴In an earlier draft, we explored the specification in which tax revenue must be spent entirely every single period; that is, $G_{j,t} = \frac{\tau_{j,t}\pi_{j,t}}{p_{jj,t}}$. We find that the results from this specification are qualitatively similar to that in which government debt is allowed.

account (KA). The CA and KA for country j are:

$$CA_{j,t} = \sum_{i=1}^N [p_{ij,t}(c_{ij,t}^w + c_{ij,t}^c) - p_{ji,t}(c_{ji,t}^w + c_{ji,t}^c)] + \sum_{i=1}^N [s_{ij,t}(1 - \tau_{j,t})\pi_{j,t} - s_{ji,t}(1 - \tau_{i,t})\pi_{i,t} + r_{j,t}k_{ij,t} - r_{i,t}k_{ji,t}] \quad (20)$$

$$KA_{j,t} = \sum_{i=1}^N [p_{jj,t}I_{ji,t} - p_{ii,t}I_{ij,t}] + B_{j,t+1}^g - B_{j,t}^g. \quad (21)$$

The current account is the sum of the trade balance, net income from dividends, and net income from factor payments for capitals. The capital account is a record of foreign capital inflows and outflows to foreign countries.

The BoP conditions, along with the budget constraints for households, imply the national income identity for country j :

$$Y_{j,t} - \frac{B_{j,t}^g r_{j,t}^b}{p_{jj,t}} = c_{jj,t}^w + c_{jj,t}^c + \sum_{i \neq j} \theta_{ij,t}(c_{ij,t}^w + c_{ij,t}^c) + I_{jj,t} + \sum_{i \neq j} I_{ji,t} + G_{j,t}. \quad (22)$$

3 Parameterization

This section discusses our choice of parameter values for a two-country version of the model specified in Section 2.¹⁵ Most of the values are taken from the literature, while country-specific parameters for the “North” (N) and the “South” (S) countries are calibrated using Canadian and U.S. data, respectively. We use the North’s price index ($P_{N,t}$) to deflate nominal values.¹⁶ Table 1 summarizes all parameter values.¹⁷

[Table 1 Here]

¹⁵In Appendix D, we explore the implications under a three-country version of the model. We find our results to be consistent across the two-country and the three-country versions of the model.

¹⁶Price index normalization is chosen over individual goods’ price normalization since as shown by [Gabszewicz and Vial \(1972\)](#), the choice of a numeraire good sold in imperfectly competitive markets is not innocuous in the general equilibrium framework. [Böhm \(1994\)](#) provides examples of how different price normalization rules influence the equilibria in an oligopolistic setting.

¹⁷We also provide a comparison between selected moments generated by the model and those in the data for the United States and Canada in the Appendix. Overall, we find that under our calibration, the model fits the data reasonably well.

The parameters characterizing the utility function and production technologies are identical across countries, except for the ones that govern the aggregate productivity processes. We set the quarterly discount rate at $\beta = 0.98^{1/4}$. We note that despite the common subjective discount factor, the steady-state rental rates of our model are heterogeneous across countries as they depend on the country's corporate tax rate.¹⁸ The degree of relative risk aversion (ψ) is set to 1, which implies an inelastic labor supply for the worker type households. The inverse Frisch elasticity (ξ) is chosen such that the steady-state labor of the worker household equals 1/3. We set elasticity of substitution (σ) to 2.66, a value that is consistent with price markups in the United States.¹⁹ Turning to the parameters governing the production function, we set $\alpha = 1/3$, a relatively standard value for capital share, and κ is set such that the Southern capitalist household's steady-state labor equals 1/3. To model the productivity processes of the two countries, we follow [Baxter and Crucini \(1995\)](#) and assume that $A_{N,t}$ and $A_{S,t}$ for the North (Canada) and the South (the United States) evolve as follows

$$\begin{bmatrix} \log A_{S,t} \\ \log A_{N,t} \end{bmatrix} = \begin{bmatrix} 0.796 & 0.131 \\ 0.000 & 0.989 \end{bmatrix} \begin{bmatrix} \log A_{S,t-1} \\ \log A_{N,t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_{S,t} \\ \varepsilon_{N,t} \end{bmatrix}, \quad (23)$$

in which $\varepsilon_{S,t}$ and $\varepsilon_{N,t}$ are assumed to be correlated with $\rho(\varepsilon_{S,t}, \varepsilon_{N,t}) = 0.434$.

The steady-state values for the tax rates are set to the average corporate statutory tax rates of Canada ($\bar{\tau}_N$) and the United States ($\bar{\tau}_S$) for the period 1981-2018. Since changes in corporate tax rates are typically persistent, with an average duration from 3 years to 9 years in the data, we set the persistence of corporate tax processes (ρ_τ) for these two countries to 0.96.²⁰ We set the steady-state government foreign debt position ($\bar{B}_j$) to match the ratio of debt payments ($r_j^b B_j^g$) to nominal GDP ($p_{jj} Y_j$) over the period 1981-2016. For the United States, this ratio averages 2.18% over that period.

Following [Bayer and Tjaden \(2016\)](#), we choose the domestic capital adjustment cost for the

¹⁸In the steady state, the gross rental rate (r_j) equals $R_j \Omega_j$ and the *net* rental rate ($\frac{r_j}{p_j}$) is $\frac{\sigma-1}{\sigma} R_j$. The term R_j is:

$$R_j = (\beta^{-1} - (1 - \delta)) \left(\frac{\sigma \alpha}{(\sigma - 1)\alpha + (1 - s)(1 - \tau_j)} \right).$$

¹⁹See, for example, [Basu \(2018\)](#), [Basu \(2019\)](#), and [Kopp et al. \(2019\)](#). This value of $\sigma = 2.66$ implies a markup of around 60%, which is consistent with recent evidence from [Kopp et al. \(2019\)](#).

²⁰Using corporate tax data from the OECD, we estimate an AR(1) fixed effect panel regression (specification: $\tau_{i,t} = \alpha_i + \rho_\tau \tau_{i,t-1} + \varepsilon_{i,t}$) with data described under Figure C.3 in the appendix. This estimation yields a quarterly persistence of 0.96 with a heteroskedasticity and autocorrelation (HAC) robust standard error of 0.011.

South (the United States) η_{SS} to equal 1.5 and, for the benchmark model, choose an equal value for the North. While there is no consensus in terms of estimates for cross-border investment costs, the literature does provide several estimates of the markups associated with international portfolio diversification. For example, [Christelis and Georgarakos \(2013\)](#) estimate the cost of investing in international stocks to be roughly 3% of the amount invested. [Sercu and Vanpée \(2008\)](#) find estimates ranging from 0.05% to as large as 27% of the implied costs of investing internationally to be dependent on both the country of origin and the target country. For the benchmark model, we set the markup for cross-border adjustment costs at 10% over the cost incurred by domestic investors; i.e, $\eta_{ij} = 1.1 \times \eta_{ii}$.

Estimates of cross-border barriers for transporting goods vary in the literature. [He et al. \(2017\)](#) and Statistics Canada provide an estimate in the vicinity of 8.1%.²¹ Their estimate, however, is not inclusive of tariffs and unobserved costs associated with transporting goods across borders. [Irarrazabal et al. \(2015\)](#) find the additive border barrier costs to be 15% on average. [Anderson and van Wincoop \(2003\)](#) estimate the additive burden on transporting goods between the United States and Canada using a gravity equation and find this cost to be roughly 31% value of the goods going from the United States to Canada. We follow [Anderson and van Wincoop \(2003\)](#) and set the iceberg trade cost from the South to North at $\theta_{NS} = 1.31$. The remaining iceberg trade cost (θ_{SN}) is set to 1.62 to achieve balanced trade in the steady state.²²

4 Dynamic Responses to a Corporate Income Tax Cut

This section examines the short-run dynamics of the two-country model following a unilateral corporate income tax cut in the South. The model parameters are outlined in Section 3 and summarized in Table 1. After analyzing this benchmark model, we consider two alternative scenarios: (i) significantly reducing the size of equity income and (ii) transfer the entire corporate tax revenue to the worker households. Our focus on the responses to corporate tax cuts, rather than tax

²¹Data are from “Trucking Across the Border: The Relative Cost of Cross-border and Domestic Trucking, 2004 to 2009.” <https://www150.statcan.gc.ca/n1/pub/11f0027m/2012081/part-partie1-eng.html> Accessed in December 2018.

²²The United States has a consistent trade deficit with Canada. Hence, a steep value for θ_{SN} is required to force a zero value for net exports (balanced trade).

increases, is motivated by the declining trend of corporate tax rates in the OECD.²³

4.1 The Effects of Corporate Income Tax Cuts

We initiate a negative shock to the innovation ($\varepsilon_{S,t}^T$) of the South's corporate tax process and calculate the impulse responses, as percentage deviations from the steady states, of selected variables. In terms of magnitude, this negative shock translates into a 14-percentage-point (pp) reduction in corporate tax on impact. This exercise is motivated by the CIT reduction enacted in the Tax Cuts and Jobs Act of 2017, in which the United States reduced its statutory corporate tax rate from 35% to 21%. Figures 2, and 3 contain the impulse response plots of selected variables.

[Figure 2 Here]

A key result of the model is that a decrease in the corporate income tax in the South implies higher income inequality in both countries.²⁴ Figure 2 presents the impulse responses of net income (gross income minus investments and adjustment costs) for all agents in the model.²⁵ Following a negative shock to the South's corporate tax rate, net income for capitalists increases immediately and remains above its steady-state levels throughout the entire horizon (40 quarters). In contrast, the Southern worker's income response falters initially and slowly becomes positive after 15 periods. The Northern worker's wage exhibits little to no change following the CIT cut in the South. Therefore, unlike the Northern capitalist, the Northern worker experiences a limited spillover effect on their income. In both countries, the increase in income inequality is at its highest upon the impact of the CIT cut, though, the effect is significantly more pronounced in the South as the income responses of the capitalist and the worker move in opposite directions.

[Figure 3 Here]

The CIT cut triggers a series of cascading effects that ultimately increase income inequality, with the reallocation of resources to increase investments into the Southern firm being the main

²³To economize on space we leave the corresponding analysis of the dynamic responses to productivity shocks in Appendix B.

²⁴In Appendix C, we consider the possibility of concurrent corporate tax competition from the Northern economy. Overall, we find that the introduction of tax competition amplifies the inequality in the income responses across the capitalists and the workers.

²⁵For the worker type households gross and net income are equal since, by construction, they have no investments.

driver. Intuitively, the tax cut makes investing in the Southern firm more attractive as its expected after-tax profits soar. Capitalists, regardless of their country of residence, ramp up their investments into the Southern firm, thereby shifting resources from the North to the South. The shift in capital supply to the South puts downward pressure on the rental rate that is sustained for about 10 periods as the capital stock builds up. The pressure produces an inverse U shape on the South’s rental rate response. We note that even though the rental rate decreases in the South, the equity income, which consists of after-tax profits and capital factor payments, of all capitalists increases. The reason is that the tax cut increases after-tax profits distribution by more than enough to offset the decrease in capital factor payments from the South. Thus, to capture the “whole” burden of corporate taxes on capital in our model with monopolistic competition, we have to consider the evolution of equity income rather than just the capital rental rate.

The South’s reduction in capital costs forces its firm to substitute labor with capital, which leads to a decrease in wages over the short run.²⁶ Due to the monopolistic competition market structure, firms in our model carry a two-step optimization problem, of which cost minimization is the first step. Observing the decrease in the rental rate, the Southern firm shifts its optimal allocation of inputs in favor of having additional capital. The decrease in labor demand results in the initial decrease in wages for the workers and capitalists alike. However, since labor income is the only source of earning for worker households, their income decreases.²⁷

As the initial drive in investment to the South stabilizes and capital stock builds up, wages in the South start an upward trajectory and revert the initial negative impact of the CIT cut. However, the speed at which wages recuperate is much slower for the worker. While it takes about 15 periods for the worker’s wage to return to its steady-state after the initial setback, the Southern capitalist’s wage setback lasts only for 6 periods. This is attributable to a reduction in the capitalist’s incentive to supply labor due to their increase in equity income, which is sufficient to overcome the fall in labor income, thereby requiring a higher wage to supply labor. Such equity and labor income dynamics for the Southern capitalist are consistent with recent empirical evidence in the United States. In particular, [Nallareddy et al. \(2018\)](#) find that following a state corporate

²⁶There are similar dynamics at play in the Northern country but in the opposite direction due to a flight of capital that increases its rental rate.

²⁷It is helpful to note that in the benchmark model workers’ labor supply is inelastic and therefore, the dynamics of their wages are the sole drivers of observed changes to their income.

tax cut, taxpayers with over \$200,000 in income see an increase in equity income and a reduction of labor income.

The dynamic aspect of our model allows us to highlight the disconnection between the short-to-medium-run and the long-run effects on the worker's income. In comparison, previous literature typically uses a static analysis that predicts increases in domestic wages following a CIT cut.²⁸ Our model shows a similar result but only after a period of adjustment. During this adjustment period, the initial effects on domestic wages, and worker's income, are negative. Such initial dynamics are driven by cross-country resource reallocation that forces firms to readjust their labor to output ratios as they factor price changes following the CIT cut. After the initial adjustment period, output in the Southern country expands along with its capital stock, which leads to an increase in wages. Previous literature has, for the most part, overlooked this transition even as it provides a critical transition mechanism to understand the effects of CIT cuts.

We continue by noting that the CIT cut also has an impact on national output, in which the South's output increases at the expense of the North. The output dynamics are driven by the increases in investment demand and, later on, by the increases in consumption demand due to lower prices. First, Figure 3 shows that capital stock and aggregate investments increase significantly for the Southern country. The increase in the capital stock is fueled by both domestic (i.e., in the South) and offshore (from the North to the South) investments. Since the consumption-investment goods are heterogeneous across the countries, the uptick in capital demand from the South is a partial driver of the South's output increase. Second, the net impact of the wages and rental rate movements is a reduction in the marginal cost of the Southern firm, which translates into lower prices. Figure 3 shows that the decrease in prices improves the South's terms of trade (ToT), therefore, all else equal, buttress the demand for the South's exports and hence the increase in output.

One feature of the model is that the South's corporate tax cut is financed by external borrowing. Thus it is not necessarily the case in our model that a CIT cut raises output, and the taxable base,

²⁸See [Gravelle \(2013\)](#) and [Auerbach \(2006\)](#) for a review of the literature.

by more than necessary to cover the windfall from decreasing the tax rate.²⁹ The South's CIT cut also has a spillover effect onto the North's government borrowing. While not a major increase as in their Southern counterpart, the North's debt position increases due to the reduction in their output due to falling investment demand and exports to the South.

[Figure 4 Here]

Finally, we investigate the welfare implications associated with a decrease in the corporate income tax rate. Similar to previous exercises in this section, we consider a tax cut of 14 percentage points, which is consistent with the reduction in the maximal corporate tax rate following the Tax Cuts and Jobs Act of 2017. Here we measure welfare as the present discounted sum of utility for both the capitalists and the workers, respectively, as follows³⁰

$$\mathbb{V}_{j,t}^c = U_t(C_{j,t}^c, L_{j,t}^c) + \beta E_t[\mathbb{V}_{j,t+1}^c], \quad (24)$$

and

$$\mathbb{V}_{j,t}^w = U_t(C_{j,t}^w, L_{j,t}^w) + \beta E_t[\mathbb{V}_{j,t+1}^w]. \quad (25)$$

Figure 4 plots the responses of welfare by capitalists and workers as percentage deviations from the steady states to a 14 percentage-point tax cut in the Southern country. These responses are consistent with the differential income responses across the two types of agents.

4.2 The Role of Equity Income

A natural question arises as to how the dynamics of the model change when firms' profits, and therefore equity income, are subdued. To answer this question, we compare the response dynamics of the baseline model with an identically calibrated model in which we increase the elasticity of

²⁹We also compare the implied tax multipliers generated by our model and from empirical evidence. Figure C.4 shows the responses of output to a corporate tax cut in the form of tax multipliers, defined as the normalized response of output following an exogenous tax decrease of one percent of output. This definition follows a recent empirical strand of the literature that studies the extent to which tax policy impacts GDP using narrative accounts of tax changes (see, for example, [Romer and Romer \(2010\)](#)). While smaller than estimates by [Romer and Romer \(2010\)](#), our estimated tax multipliers are consistent with [Favero and Giavazzi \(2009\)](#), who find tax multipliers to be either insignificantly different from zero or far smaller than one.

³⁰Our analysis of the welfare implications of a CIT cut is motivated, in part, by recent works by [Engler and Wulff \(2014\)](#) on the impact of financial liberalization on consumption and welfare across capitalists and workers.

substitution (σ) to 10.³¹ A larger value of σ reduces monopoly power which results in smaller markups over the marginal cost and all else equal, lowers firms' profits thereby reducing equity income.

[Figure 5 Here]

Figure 5 presents the impulse responses for the baseline model (solid line) and the model with $\sigma = 10$ (dashed line). Intuitively, lower profits reduce the net return of investments, especially for offshore investments as their adjustment costs are higher than domestic investments. Thus, the corporate tax shock still generates a re-allocation of capital to the South but at a much lower scale than in the baseline model.

The model with the subdued equity income channel ($\sigma = 10$) thus exhibits a smaller increase in income inequality. For the Southern worker, the initial income losses, as well as the latter gains, are smaller than under the baseline specification. Capitalists' income increases with a much smaller magnitude than under the baseline model. Combined, these responses show that when the equity income channel is subdued, a corporate tax cut results in a less pronounced increase in income inequality than under the baseline specification.

4.3 The Role of Transfers

In the baseline model, we have assumed that corporate tax revenues are used as government consumption and therefore are purely wasteful. In this section, we relax this assumption and allow such revenues to be distributed as transfers to the workers. In particular, workers receive transfers equivalent to the amount of tax revenue collected $\tau_{j,t}\pi_{j,t}$, which also implies no government borrowing. We next compare the responses under our baseline cases with one in which the workers do receive the transfers $\tau_{j,t}\pi_{j,t}$ in Figure 6.

[Figure 6 Here]

Intuitively, we expect the model with tax transfers to amplify the magnitude of the workers' income responses due to the additional source of income reduction. Figure 6 shows that allowing tax revenues to be transferred to the workers amplifies the income responses of all agents; furthermore,

³¹The shock is identical to the one in the previous subsection: a 14 pp decrease in the South's corporate tax rate.

the negative impact on the Southern worker becomes significantly more pronounced and does not quickly revert to its previous steady-state level. Most importantly, the model with tax transfers generates significantly a larger increase in income inequality following a CIT cut.

5 Extensions

This section reports additional results that alter our benchmark analysis in several ways to check on their sensitivity and our interpretations of the findings. We start with a two-country version of the model and use the parameterization in Section 3 as the baseline.

5.1 High Cross-border Investment Costs

Since one key channel through which a CIT cut can impact income re-allocation across the two countries is through cross-country investment, we recalculate our impulse responses using a higher value for cross-border investment markups. Specifically, we increase the cross-border investment markup to invest in the South (η_{SN}) by 50%. We next plot selected responses to an unexpected decrease in corporate tax at the South (14 percentage points) in Figure 7. Here we overlay the responses under the baseline (solid lines) and under the case with the increase in cross-border investment markups (dashed lines).

[Figure 7 Here]

The exercise presented in this section shows that increased frictions to investments into the South significantly impact the dynamics *in the North* via capital market disruptions, even if investment frictions into the North remain constant. Naturally, the increased frictions reduce the magnitude of foreign investments into the South following its CIT cut. Surprisingly, even though the financial friction associated with investing in the North remains constant, the outflow of capital from South to North switches from positive, in the baseline model, to negative. While at first glance, this result seems odd, it is attributable to capital crowding. Relative to the baseline case, the higher financial frictions force the reallocation of a portion of resources used to increase foreign holdings in the South into domestic investment. The additional availability of domestic capital in the North precludes the Southern capitalist from investing part of the benefits from the CIT cut

into the North.

The disruption in the capital markets also impacts labor markets and income inequality, especially for the North. The increased financial frictions to the South result in higher income inequality, relative to the baseline model, over the short-to-medium run and lower income inequality in the North after a period of adjustment. The reason behind this effect is that the increased financial friction scenario and the baseline produce almost identical responses in the Northern worker's wage. However, the Northern capitalist sees a higher income gain, relative to baseline, due to a decrease in adjustment costs incurred for adjusting their investment in addition to the increase in labor income. In the South, the worker's wage decreases further than in the baseline scenario. This is due to the additional availability of domestic investment which pushes rental rates further down and encourages the firm to replace capital with labor by a higher degree than under the baseline scenario. The Southern capitalist also sees a smaller increase in their net income due to reduced labor income, capital factor payments. Thus, we observe relatively higher income inequality as a result of the CIT cut in the scenario with higher financial frictions.

As the barrier to capital mobility increases, immobile labor (particularly for worker-type households) bears a smaller portion of the corporate tax burden and thus is less likely to benefit from a CIT cut. As capital is less likely to be shifted abroad (Figure 7), labor bears a relatively smaller burden of the tax since the resulting larger domestic capital increases marginal productivity, which increases wages. The workers thus are less likely to benefit from a tax cut in this case since the burden on labor is smaller to start with. From a policy perspective, this implies that a CIT cut combined with financial liberalization (i.e., higher capital mobility) can help alleviate income inequality as the cut is more beneficial for the workers than for the capitalists. Our result is largely consistent, in the short run, with the large and growing literature on the effects of corporate tax incidence in open economy settings (see [Gravelle \(2013\)](#) for a review of this literature). One key finding of this literature is that less capital mobility implies a smaller tax burden to labor relative to capital. Any reduction in the corporate tax burden, following from this literature, would help better alleviate income inequality under increasing financial liberalization (i.e., a smaller barrier to capital mobility).

5.2 High Cross-border Trade Costs

We next examine how higher trade costs impact income responses of capitalists and workers to a decrease in corporate tax. To that end, we increase the cross-border trade costs by 15% and plot, in Figure 8, the impulse responses of selected variables for capitalists and workers to a decrease in corporate tax (14 percentage points) in the South.

[Figure 8 Here]

While the CIT cut makes investing in the Southern firm more attractive, higher trade costs dampen firms' profits as exports decrease relative to the baseline model. These two contrasting channels imply that the stimulating effects of a corporate tax cut can potentially be negated by increases in trade costs. Moreover, although the scenario with higher trade costs results in smaller income gains for the capitalists, it also leads to a greater negative impact on workers' wages and income. Because all agents are all worse-off, the increased trade scenario yields almost identical responses in income inequality relative to the baseline model. From a policy perspective, trade liberalization (i.e., lower trade costs) can lift all agents' income without any significant impact on income inequality, as evidenced from Figure 8.

5.3 Adding a Non-tradable Service Sector

In this section, we augment the benchmark model in section 2 with a “service occupations” sector.³² Our exercise is motivated by recent evidence by [Autor and Dorn \(2013\)](#), who document that wages of “low” skill workers in service occupations have fared better than wages of workers that belong to the middle portion of the skill spectrum. Examples of service occupations are jobs that assist or care for others (e.g., hairdressers, caregivers, food services, janitors, etc.).

In each country, a “service” worker (denoted by the superscript s) provides manual labor to produce a single composite good consumed by the domestic households. The utility function of the agents is a nested CES function with the top layer composed of the composite consumption

³²We are especially grateful to an anonymous referee for this suggestion.

good $C_{j,t}^a$ and the service good $M_{j,t}^a$ as follows

$$\max \mathbb{E} \sum_{t=1}^{\infty} \beta^t \left\{ [(C_{j,t}^a)^{\Xi} + (M_{j,t}^a)^{\Xi}]^{(1/\Xi)} - \chi \log L_{j,t}^a \right\} \quad \text{for } a = w, c, s \quad (26)$$

Similarly to the “worker” household, the service workers’ income consists of labor income only:

$$\omega_{j,t}^s L_{j,t}^s = \sum_{i=1}^N p_{ji,t} c_{ji,t}^s + P_{j,t}^m M_{j,t}^s.$$

Unlike the consumption goods, the service output is sold in a perfectly competitive market, at price $P_{j,t}^m$, and is not subject to the corporate tax. The production function is a linear technology with only labor as an input:

$$Y_{j,t}^m = \mu_j L_{j,t}^s. \quad (27)$$

Furthermore, the nature of the service occupations implies that the service output is consumed by domestic agents only. Let $M_{j,t}^a$ represent the demand of agent a for services, then the market-clearing condition is:

$$Y_{j,t}^m = M_{j,t}^w + M_{j,t}^c + M_{j,t}^s. \quad (28)$$

We plot the selected impulse responses to a 14-percentage-point negative shock to corporate income tax in Figure 9. Income for the capitalists in both countries increases while income for the workers (traditional and non-corporate service alike) either decreases or largely stays the same following a tax cut arising from the South. Here traditional workers are simply denoted as workers in the figure since they function similarly to the workers in the benchmark model. This result is consistent with the result obtained under the benchmark model in which a corporate tax cut amplifies the income asymmetry between the capitalists and the workers.

[Figure 9 Here]

One notable result from Figure 9 is the discrepancy in income responses across the traditional and service workers. In particular, service workers in the North are more likely to benefit from the tax cut than their traditional counterparts. On the other hand, in the South, service workers experience an income loss that is significantly smaller than the traditional workers on impact. We

attribute the smaller decrease in wages for the service workers to the fact that they do not face a substitution of cheaper capital for their labor. There is still a decrease since the consumption good is now cheaper and thus some substitution effects are still at play. This pattern is consistent with [Autor and Dorn \(2013\)](#), who document, using U.S. data from 1980 to 2005, that workers in the middle of the skill spectrum (which corresponds to the traditional workers in our case) experience lower income growth than workers that belong to the lowest part of the skill spectrum.

6 The Long-run Impact of A Corporate Tax Cut

This section examines the long-run impact of a corporate tax cut in our model. We do so by repeatedly solving for the steady-state values of capital, and output, as well as income of the capitalist and the worker by changing the steady-state corporate tax rate in the South (i.e., the United States) while keeping all other parameters as in Section 3. We plot these steady-state values in Figure 10.

[Figure 10 Here]

A corporate tax cut in the South increases the income of the capitalists in both countries and the Southern worker. The worker in the South, where the tax cut was initiated, observes an increase in their income due to higher labor demand as a result of the higher output that also increases their wages. Thus, unlike under the short-run dynamics, here the Southern worker benefits from the CIT cut. On the other hand, the capitalists' long-run responses are consistent with the short-run effects, as evidenced by the impulse responses in section 4, in the sense that their income increases regardless of where the tax cut takes place.

While the South's capitalist and worker both experience income gains as CIT decreases, the capitalist's gains significantly outpace those of the worker; hence an increase in income inequality. This result is consistent with the negative correlation between income inequality and the average corporate tax burden in Figure 11, which plots the GINI coefficients against the ratio of corporate tax over GDP for multiple developed economies.

[Figure 11 Here]

There are also spillover effects into the North’s output and capital, which translates into a negative impact on the Northern worker. A decrease in the South’s CIT rate results in a higher level of capital stock and consequently, higher output. The South’s output and capital expansions come at the expense of the North, which experiences a decrease in its capital stock and output. Such contraction in the North’s output then reduces labor demand, which reduces wages. Moreover, while the Northern capitalist can offset the decrease in their labor income, the Northern worker experiences a decrease in their income even though no policy change was implemented in the North.

7 Conclusion

The Tax Cuts and Jobs Act of 2017, which significantly reduces the corporate tax burden in the United States, is another example of the overall decline in corporate tax across OECD economies since the early 1980s. This trend, along with the overall increase in income inequality, has re-kindled an interest in studying the impact of corporate tax cuts on the relative income of different agents in the economy. Our paper contributes to this literature by examining the cross-country implications of corporate tax cuts on income inequality in a tractable multi-country DSGE model.

Calibrating a two-country version of the model to U.S. and Canadian data, we show the benefits of tax cuts significantly favor agents who own capital and those who can diversify their investment portfolios across borders. Our results are consistent with recent empirical evidence à la [Nallareddy et al. \(2018\)](#) that corporate tax cuts can significantly increase income inequality. From a policy perspective, we also find trade and cross-border financial liberalization to amplify the impact of corporate tax cuts on income inequality.

Our paper suggests that policymakers, of the country planning a CIT cut and its trading partners, need to consider a number of dilemmas. Domestically, our model predicts that a CIT would increase income inequality, an unpopular issue in political debates. More importantly, policymakers are faced with enforcing a policy that negatively impacts workers’ wages and income over the short run, but that will ultimately increase their welfare and income over the long run. From an international perspective, a CIT cut in one country has a spillover effect onto its trading partners, including an increase in inequality and a decrease in their output. These issues should be carefully considered when designing a country’s corporate tax policy.

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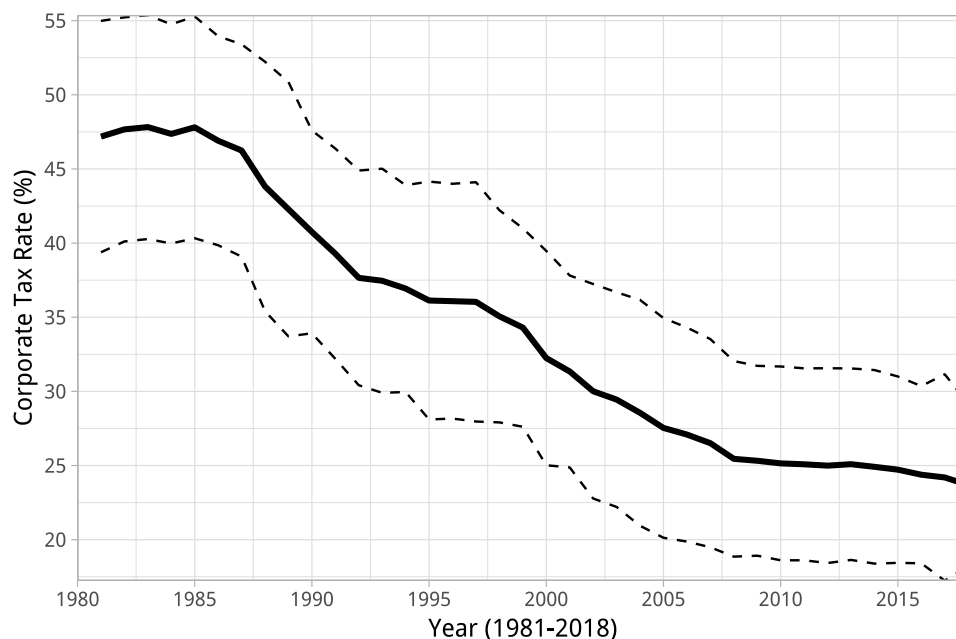
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Table 1: Parameters

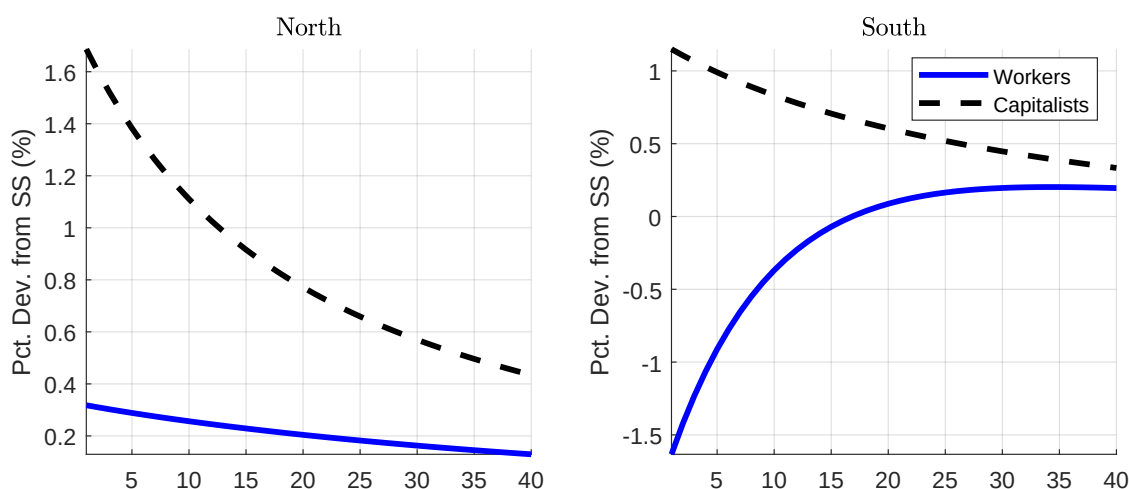
Parameters	Values	Notes	Description
<i>Utility</i>			
β	$0.98^{1/4}$	Annualized (2%)	Discount rate
ψ	1		Degree of relative risk aversion
ξ	1/3	Peterman (2016)	Inverse of Frisch elasticity
χ	1	-	Labor disutility
σ	2.66	Kopp et al. (2019)	Elasticity of substitution
<i>Production</i>			
α	0.330	Backus et al. (1992)	Capital share
κ	0.354	s.t $\bar{L}_S^c = 1/3$	Labor share for capitalists
δ	0.025	Backus et al. (1992)	Depreciation rate for capital
ρ_N^A	0.989	Baxter and Crucini (1995)	Persistence of TFP for Canada
ρ_S^A	0.796	Baxter and Crucini (1995)	Persistence of TFP for the United States
<i>Taxes</i>			
$\bar{\tau}_S$	0.36	Average CIT rate for the United States	Steady-state corporate tax rate
$\bar{\tau}_N$	0.25	Average CIT rate for Canada	Steady-state corporate tax rate
ρ_τ	0.96	Our calibration using OECD data	Persistence of CIT process
<i>Adjustments and transportation cost</i>			
θ_{NS}	1.31	Anderson and van Wincoop (2003)	Transportation cost
θ_{SN}	1.62	Balance trade in steady state	-
η_{jj}	1.5	Bayer and Tjaden (2016)	Capital adjustment cost
η_{ij}	$1.1 \times \eta_{jj}$	Sercu and Vanpée (2008)	Cross-country capital adjustment cost

Figure 1: Corporate Taxes in the OECD



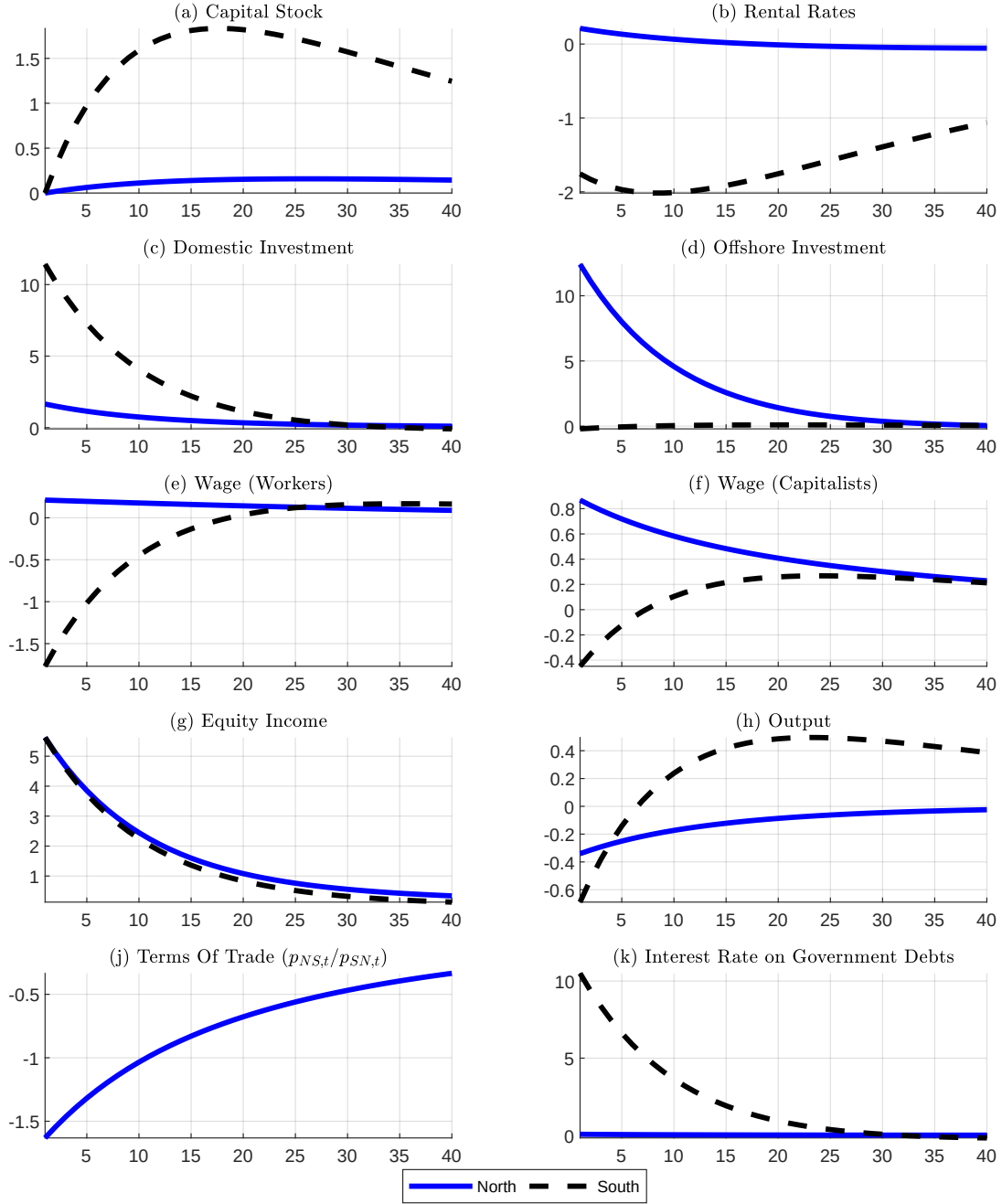
Note: Data are from the OECD's tax library database. The solid line denotes the mean across countries and the dotted lines denote the one standard deviation bands. The countries include Australia, Austria, Belgium, Canada, Chile, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Israel, Italy, Japan, Korea, Latvia, Lithuania, Luxembourg, Mexico, the Netherlands, New Zealand, Norway, Poland, Portugal, Slovak Republic, Slovenia, Spain, Sweden, Switzerland, Turkey, the United Kingdom, and the United States. We plot tax rates for these individual countries in Figure C.3 in the appendix.

Figure 2: Income Responses to a Decrease in Corporate Income Tax in the South



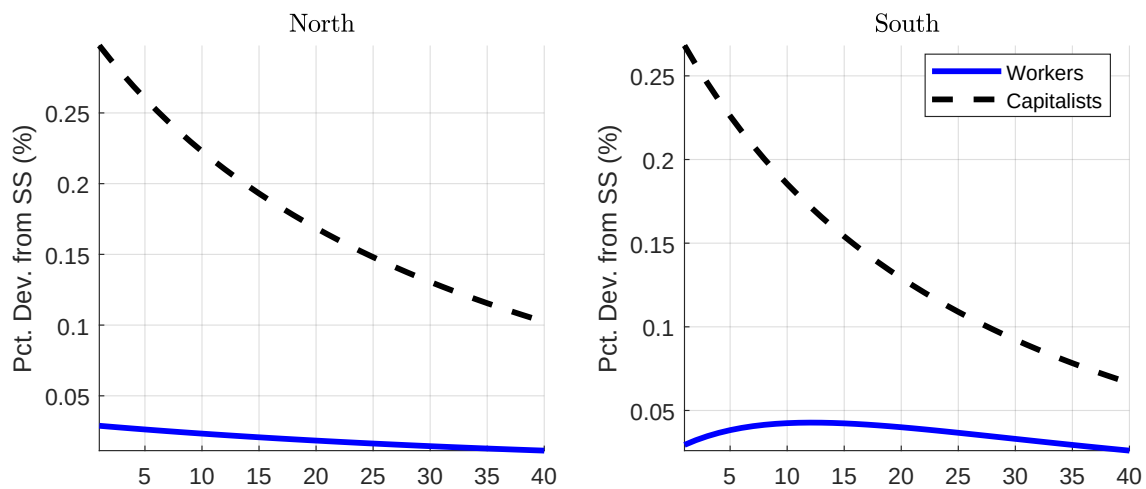
Note: This figure plots the responses to a negative shock (equivalently, 14 percentage points) to the innovation of the tax process in the South. Here income is defined as the total disposable income.

Figure 3: Responses to a Decrease in Corporate Income Tax in the South



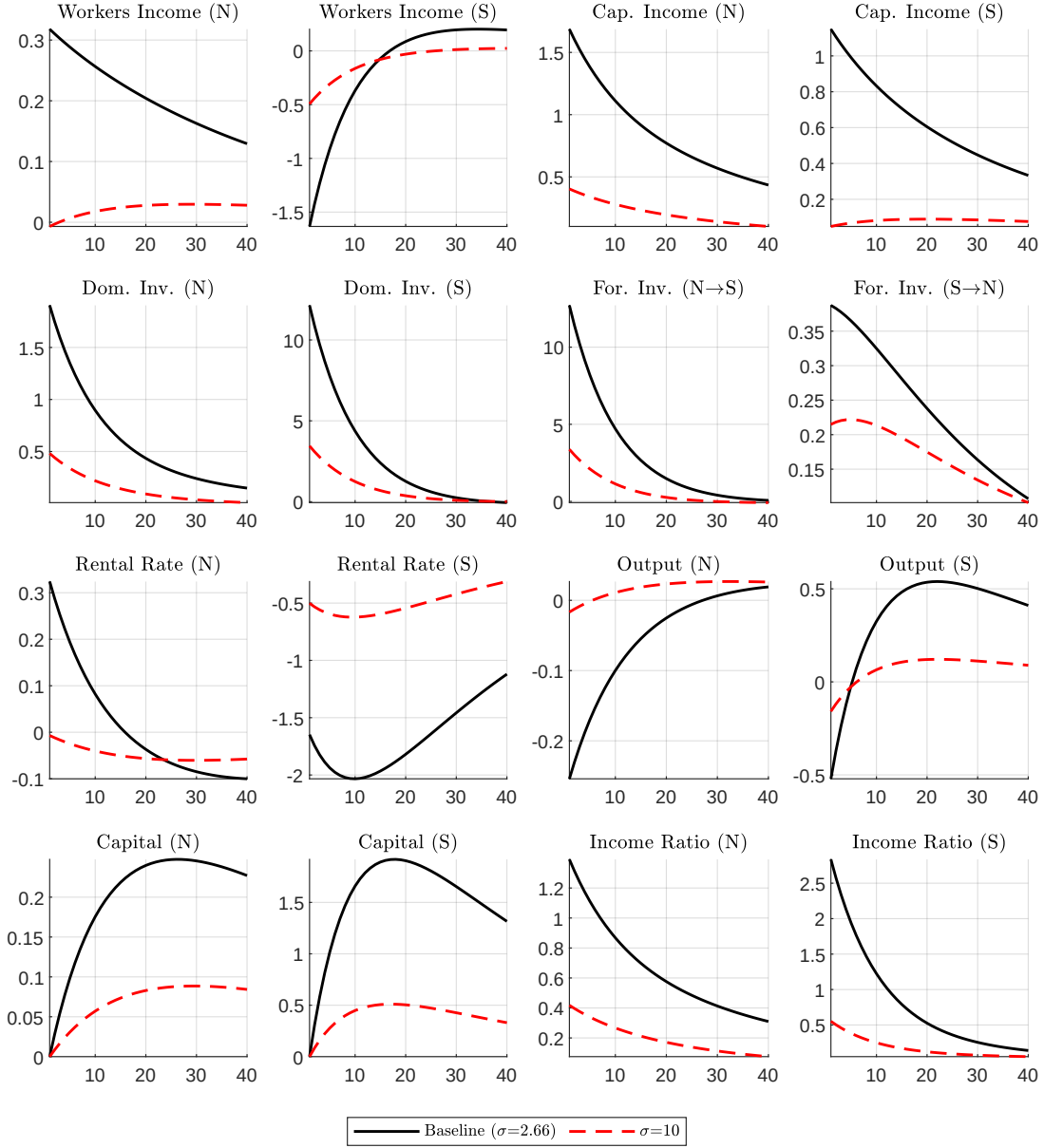
Note: This figure plots the responses to a negative shock (14 percentage points) to the innovation of the tax process in the South. The terms of trade is defined as the ratio of the South's export price to its import price. Equity income applies to only the capitalists as workers do not have equity income. The units of all subfigures are percentage deviation from the corresponding steady states.

Figure 4: Welfare Responses to a Decrease in Corporate Income Tax in the South



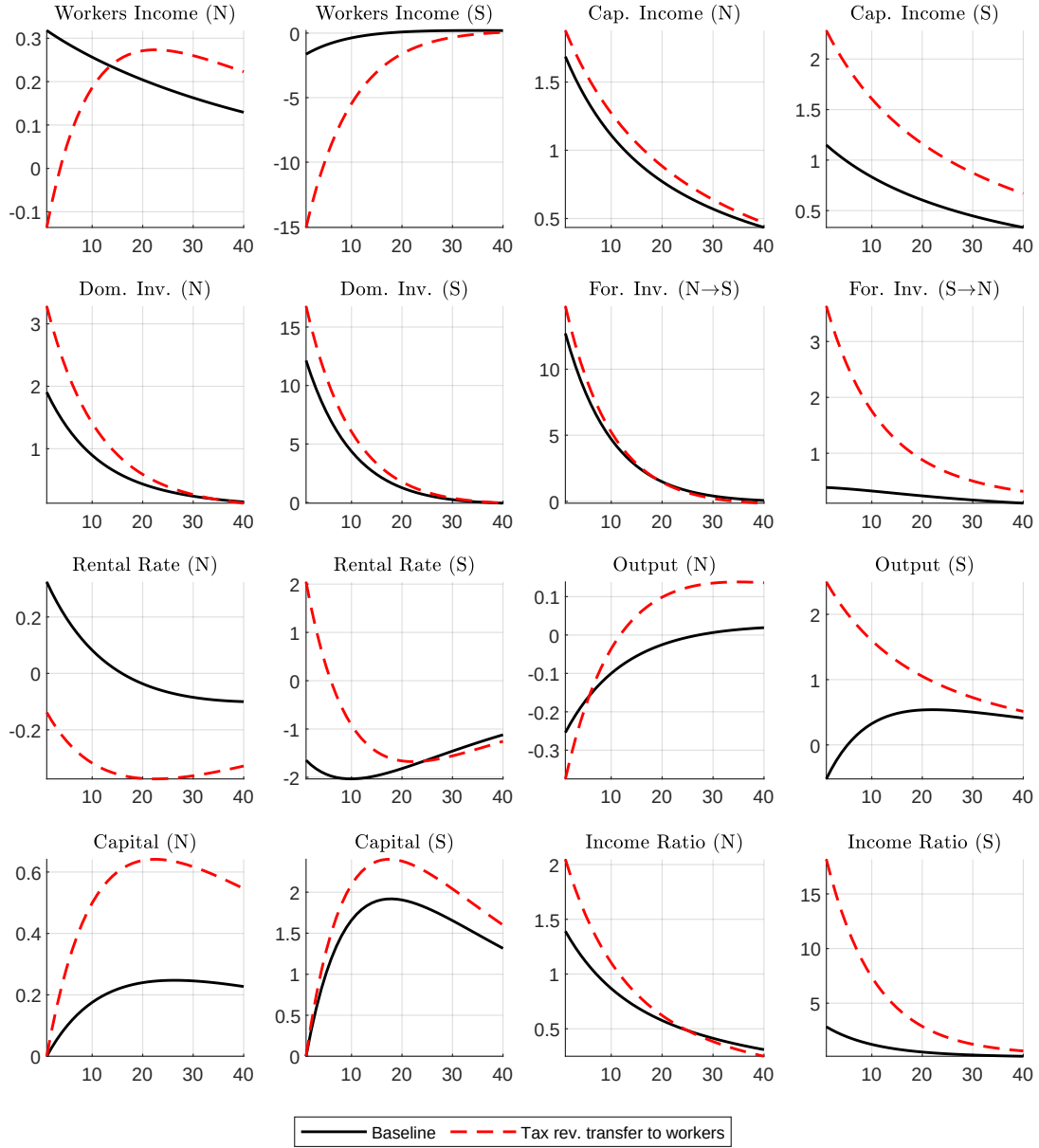
Note: This figure plots the responses to a negative shock (equivalently, 14 percentage points) to the innovation of the tax process in the South. Here income is defined as the total disposable income.

Figure 5: The Role of the Equity Income Channel



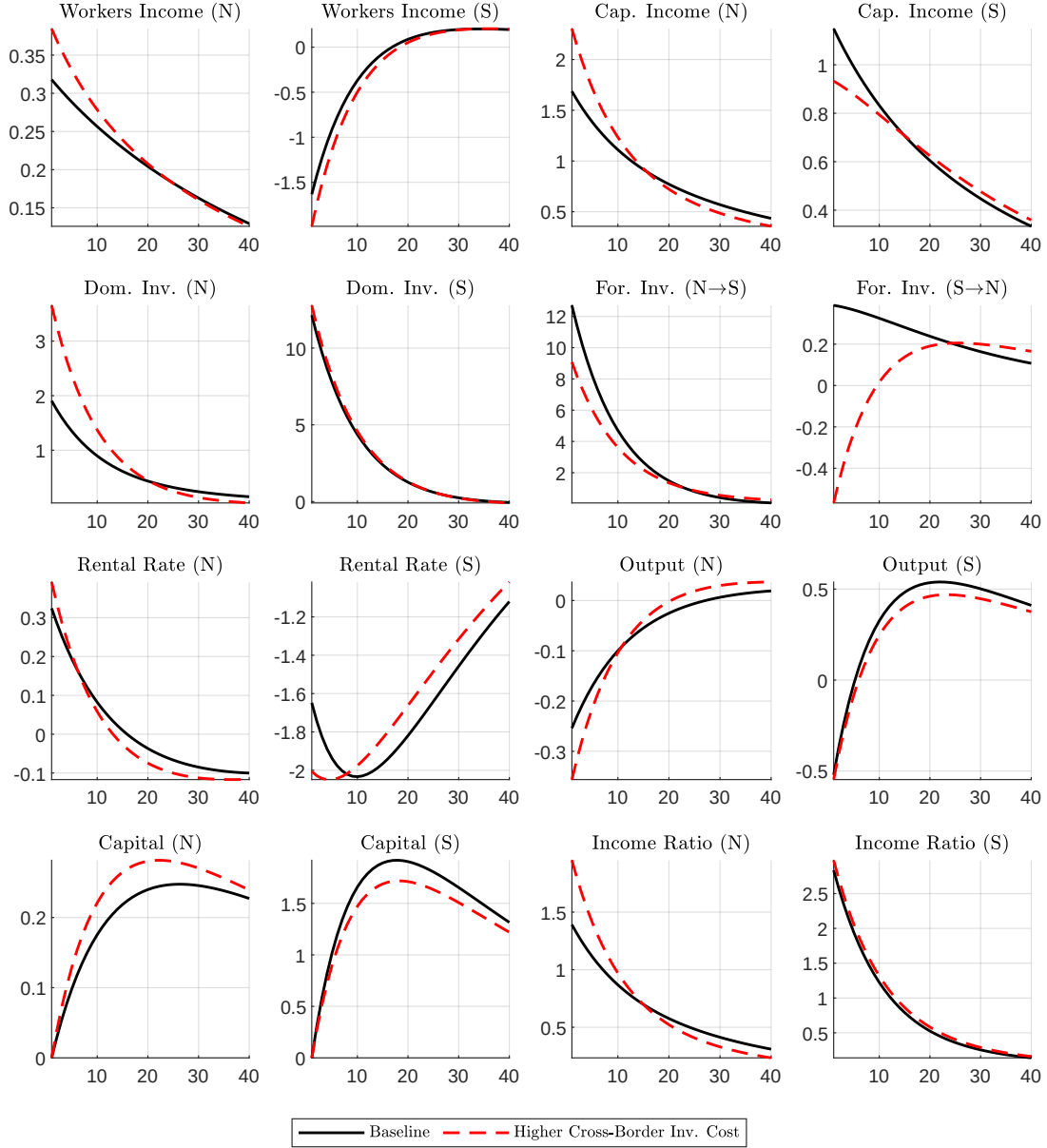
Note: This figure plots the responses to a negative shock (14 percentage points) to the innovation of the tax process in the South. In this exercise, we decrease the firms' monopoly power by increasing σ to 10 (dashed line) from the baseline calibration (solid lines). Income ratio is defined as the ratio of the capitalists' income over the workers' income.

Figure 6: The Role of Transfers



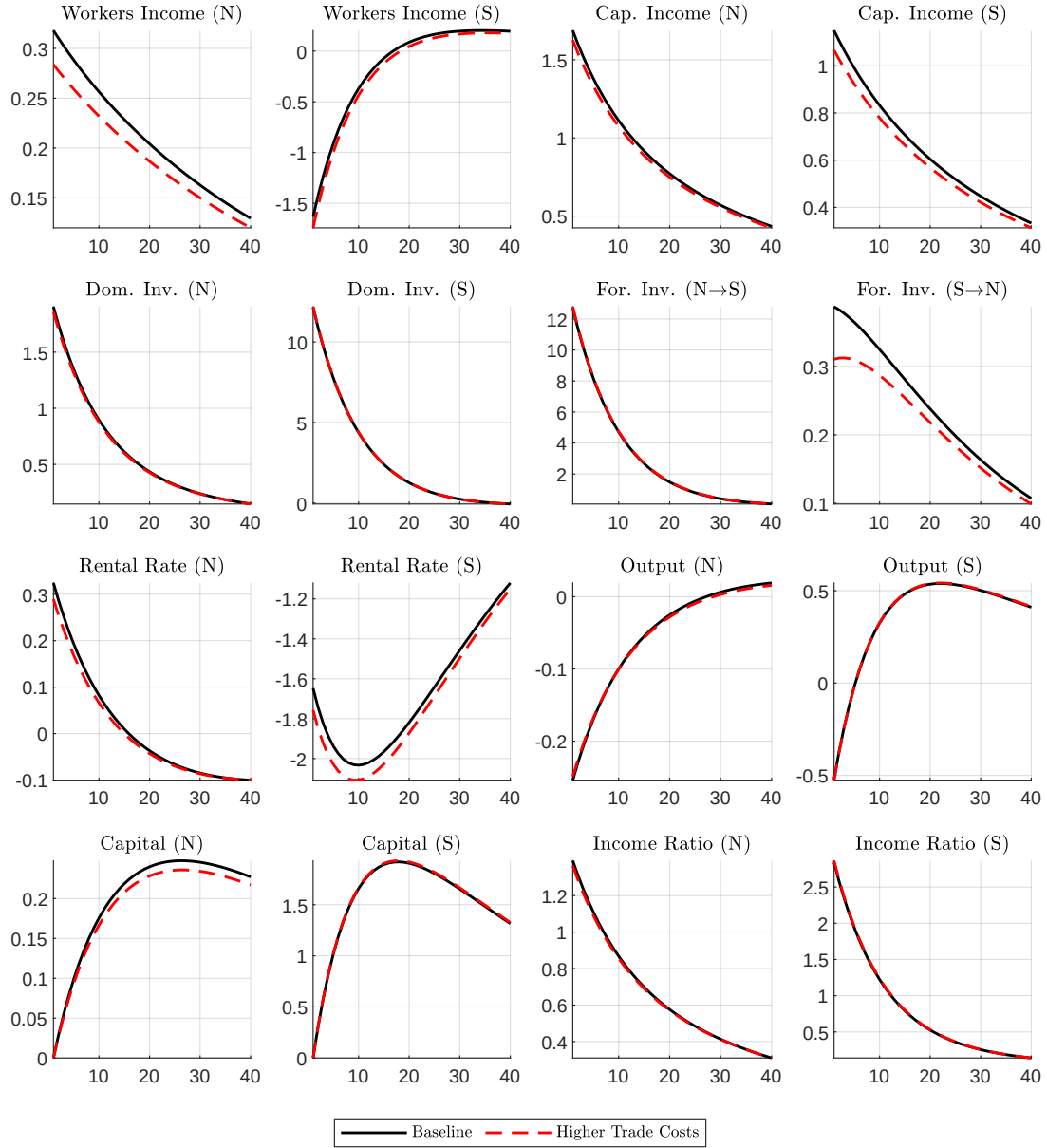
Note: This figure plots the responses to a negative shock (14 percentage points) to the innovation of the tax process in the South. In this exercise, we do not allow the workers to receive transfers (dashed line) and compare the responses to ones under the baseline calibration (solid lines). Income ratio is defined as the ratio of the capitalists' income over the workers' income.

Figure 7: High Cross-border Investment Costs



Note: This figure plots the responses to a negative shock (14 percentage points) to the innovation of the tax process in the South. In this exercise, we increase the cross-border (from North to South) investment adjustment cost η_{NS} by 50% (dashed line) from the baseline calibration (solid lines). Income ratio is defined as the ratio of the capitalists' income over the workers' income.

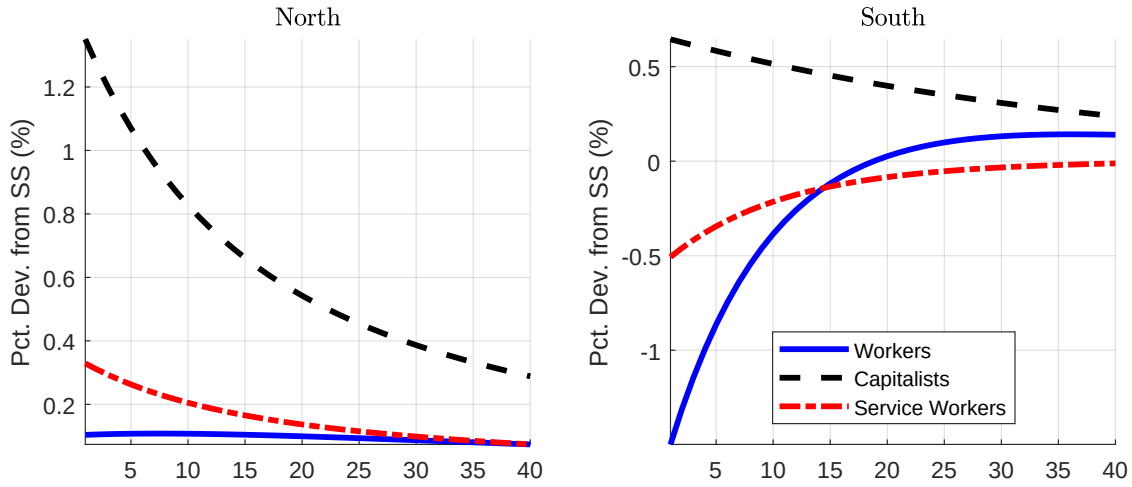
Figure 8: High Trade Costs



Note: This figure plots the responses to a negative shock (14 percentage points) to the innovation of the tax process in the South. In this exercise, we increase the cross-border trade costs θ_{ij} by 15% (dashed line) from the baseline calibration (solid lines). Income ratio is defined as the ratio of the capitalists' income over the workers' income.

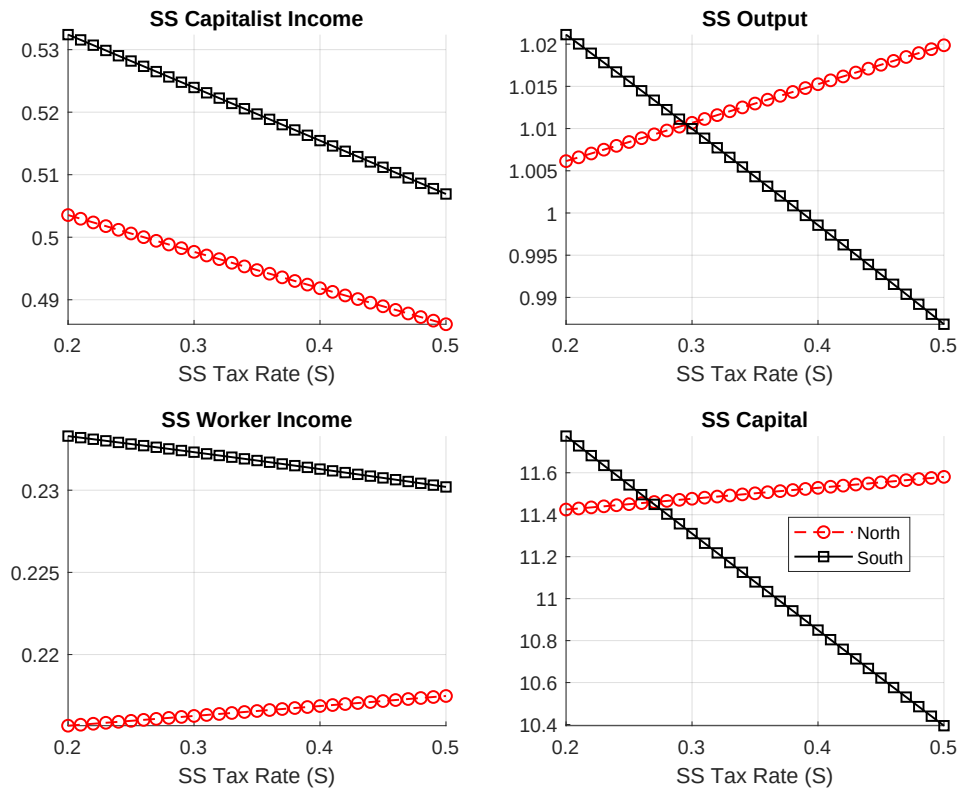
Figure 9: Extension: Adding the Service Sector

Income Responses to a Decrease in Corporate Income Tax in the South



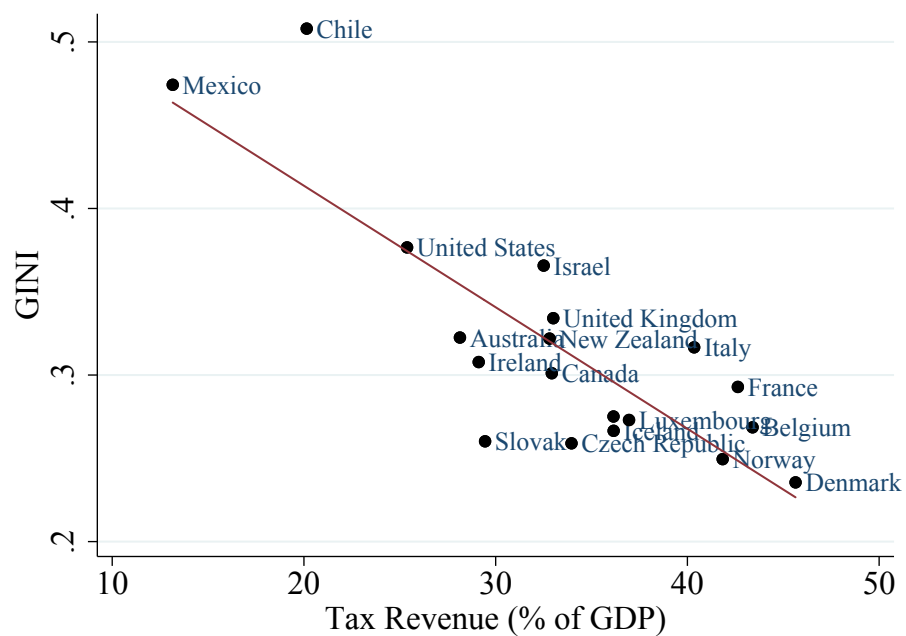
Note: Responses to a 14 percentage-point tax cut from the Southern country under the three-agent variant of the model.

Figure 10: Long-run Impacts of Corporate Income Tax Cut in the South



Note: This figure plots the steady-state values of selected variables implied by the model with respect to different U.S. corporate tax rates ($\bar{\tau}_S$).

Figure 11: Average Corporate Tax Burden and Income Inequality (1981-2018)



Note: Data are from the OECD. Corporate tax revenues are defined as a percentage of GDP. Both tax revenues and GDP are nominal and of annual frequency.

Appendix – Not for Publication

A Comparison of Selected Moments: Model and Data

We examine the extent to which our model fits with the data by comparing selected moments generated by the model and the ones from the data. To that end, we start with quarterly data for private consumption, investment, and real output for Canada and the United States from the St Louis' FRED. The data are seasonally adjusted and de-trended using the Hodrick and Prescott filter with a smoothing parameter of 1,600. Across both the model and the data, we calculate the standard deviations of output, consumption, and investment, along with the correlation between the output and consumption series in Table A.1. Overall, we find the model to be consistent with the data based on these selected moments under the baseline calibration.

Table A.1: Model versus Data

Variables	Model	Data
σ_N^Y	0.054	0.060
σ_S^Y	0.053	0.068
σ_N^I	0.033	0.031
σ_S^I	0.029	0.020
σ_N^C	0.007	0.007
σ_S^C	0.007	0.006
$\rho(C_N, Y_N)$	0.622	0.652
$\rho(C_S, Y_S)$	0.593	0.518
$\rho(C_N, C_S)$	0.404	0.461
$\rho(Y_N, Y_S)$	0.532	0.579

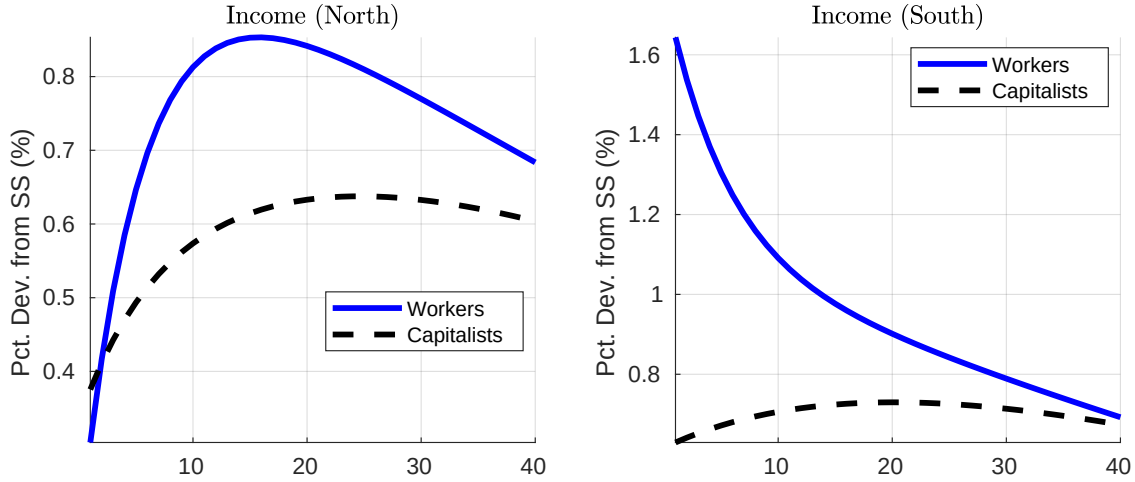
Note: The North (N) and the South (S) correspond to Canada and the United States, respectively. Data are from the St. Louis' FRED, are of quarterly frequency, and are seasonally adjusted. σ_j^Y , σ_j^I , and σ_j^C denote the standard deviation of output, investment, and consumption in country j , respectively.

B The Effects of an Increase in TFP

We next examine the extent to which agents across the two countries respond to an increase in TFP in the South. Similar to Section 4.1, our focus is on the responses to shocks from the South since the responses to the same shocks initiated from the North are qualitatively similar. Turning to more details, we initiate a one-standard-deviation shock to the TFP process of the South and calculate the responses as percentage deviations from their corresponding steady states in Figure B.2.

A positive shock to TFP has an effect similar to a CIT cut in the sense that it increases both output and capital stock. This is attributable to the fact that investing in capital in the South is more profitable given that the South is now more productive. Such a build-up is driven by an

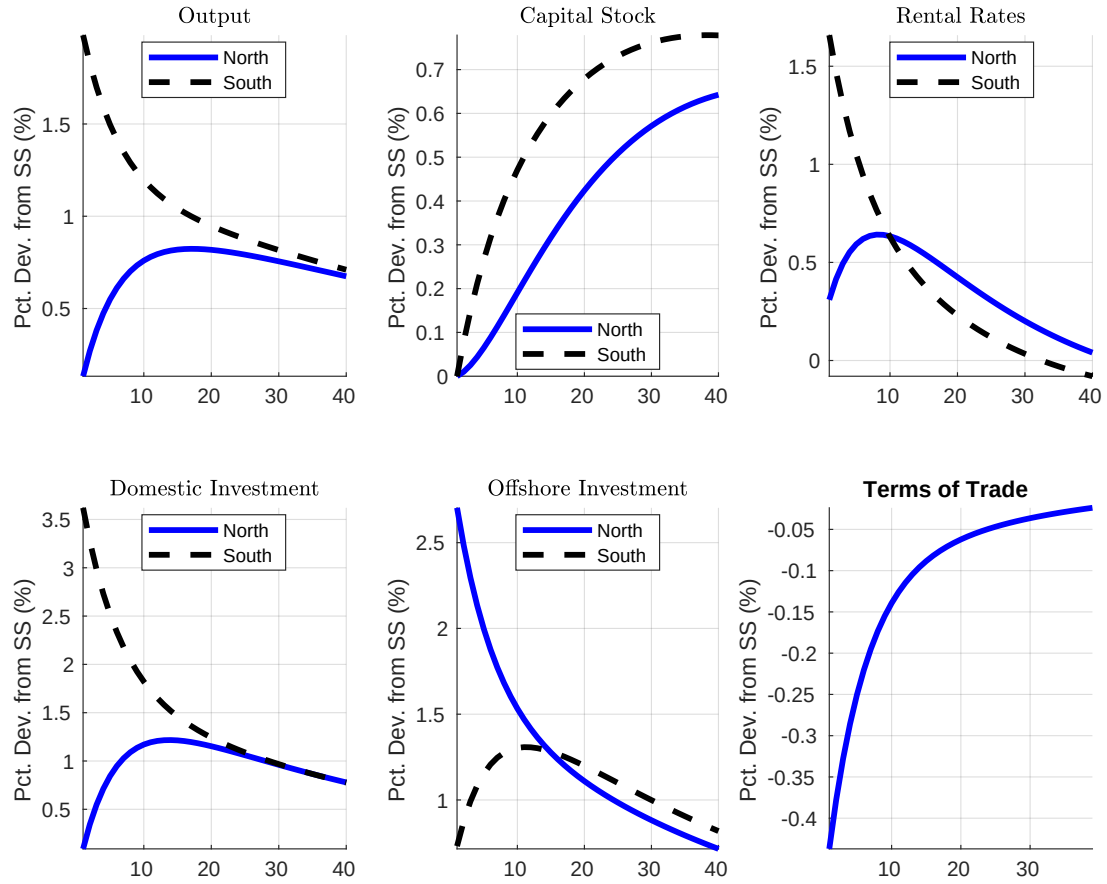
Figure B.1: Income Responses to an Increase in TFP (South)



Note: This figure plots the responses to a positive one-standard-deviation shock to the innovation of the TFP process in the South. Here income is defined as the total disposable income.

increase in the investment flow from domestic (from the perspective of the South) capitalists as well as foreign investments from capitalists in the North since capital is now more productive. This reallocation channel of capital to the South results in an initial decrease in capital stock in the North, as well as higher interest rates across the two countries as investments ramp up. Despite such reallocation, both agents from the two countries are well-positioned to benefit from the shock. Specifically, in Figure B.1, income increases significantly for both the capitalists and the workers across the two countries.

Figure B.2: Responses to an Increase in TFP (South)



Note: This figure plots the responses to a positive one-standard-deviation shock to the innovation of the TFP process in the South. Here the terms of trade is defined as the ratio of the South's export price to its import price.

C The Effects of Tax Competition

In this section, we analyze the effects of a decrease in corporate tax rate from the Southern country in the presence of competition from the Northern country. In particular, here we allow innovation to the corporate tax processes in the two countries to be positively correlated ($\text{Corr}(\varepsilon_{N,t}^\tau, \varepsilon_{S,t}^\tau) = 0.8$) with each other. This assumption reflects the idea that countries might reduce their corporate tax rate to attract and retain capital investments, which is consistent with the recent downward trend in corporate tax rates across OECD economies (as evidenced in Figure 1).

[Figure C.1 Here]

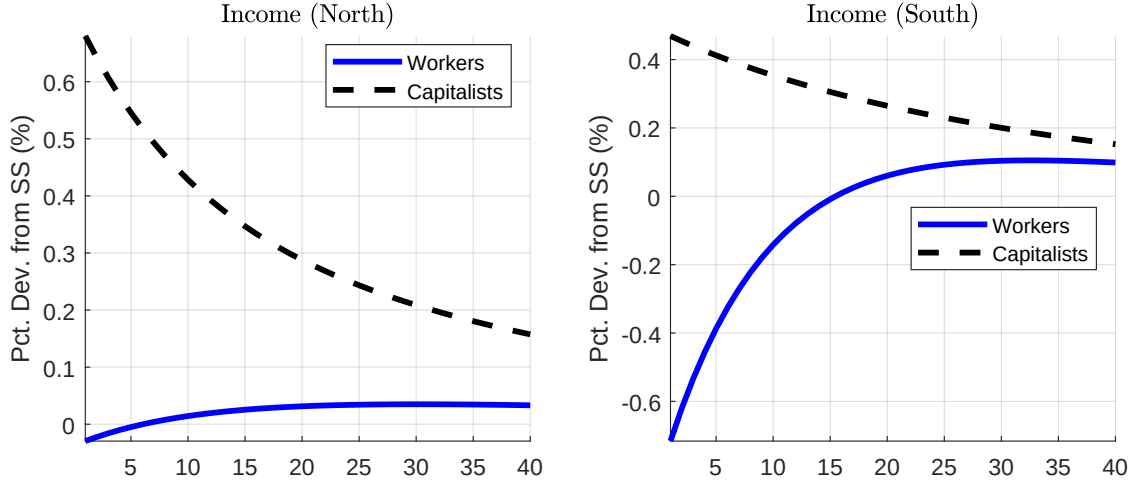
Figure C.1 shows the impulse responses to a corporate tax cut (14 percentage points) in the Southern economy that is subject to competition from the Northern country. We find our result on the differentiated responses across the two types of agents to be consistent with the responses under no tax competition in the main text. Moreover, since the Northern country also retaliates by cutting its corporate tax rate, the capitalists in both countries are even better positioned to take advantage of the double tax cuts from both the Southern and the Northern countries. For this reason, we consider our result that inequality increases following a corporate tax cut in the main text to be somewhat conservative.

[Figure C.2 Here]

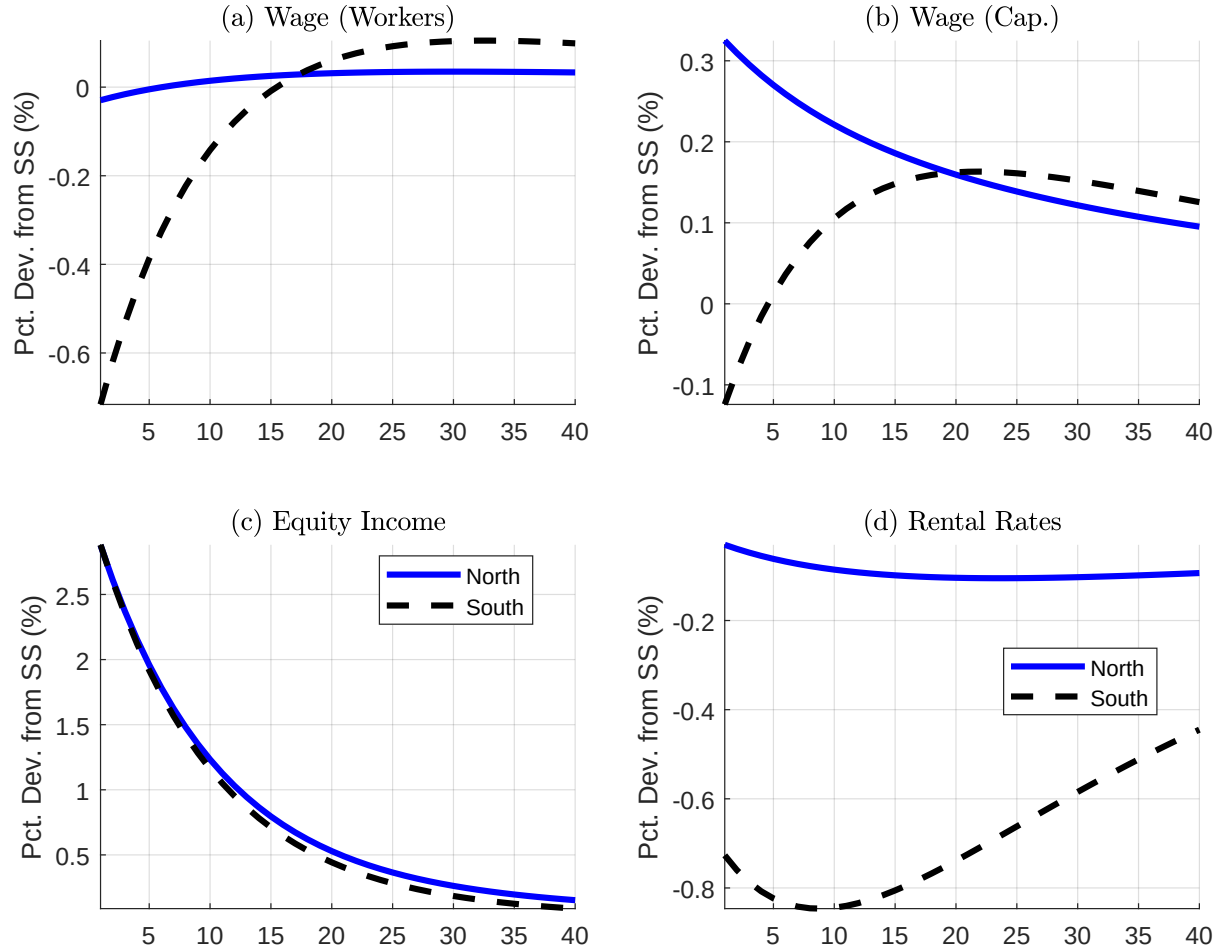
We next consider the welfare implications of a corporate tax cut under competition in Figure C.2. Here welfare is defined as in Equations 24 and 25 in the main text. Consistent with our earlier results on the distributional effects of income in Figure C.1, here we also find that capitalists are better positioned to take advantage of the tax cuts from both countries: one unilateral tax cut and another by competition.

Figure C.1: Responses to a Southern Tax Cut - with Retaliation from the Northern Country

(i) Aggregate Income

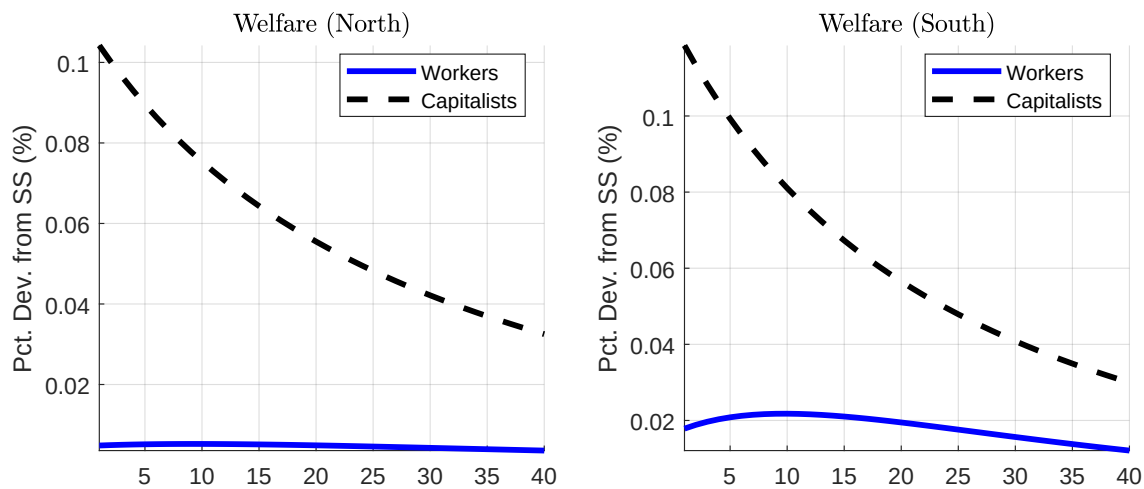


(ii) Various Components of Income



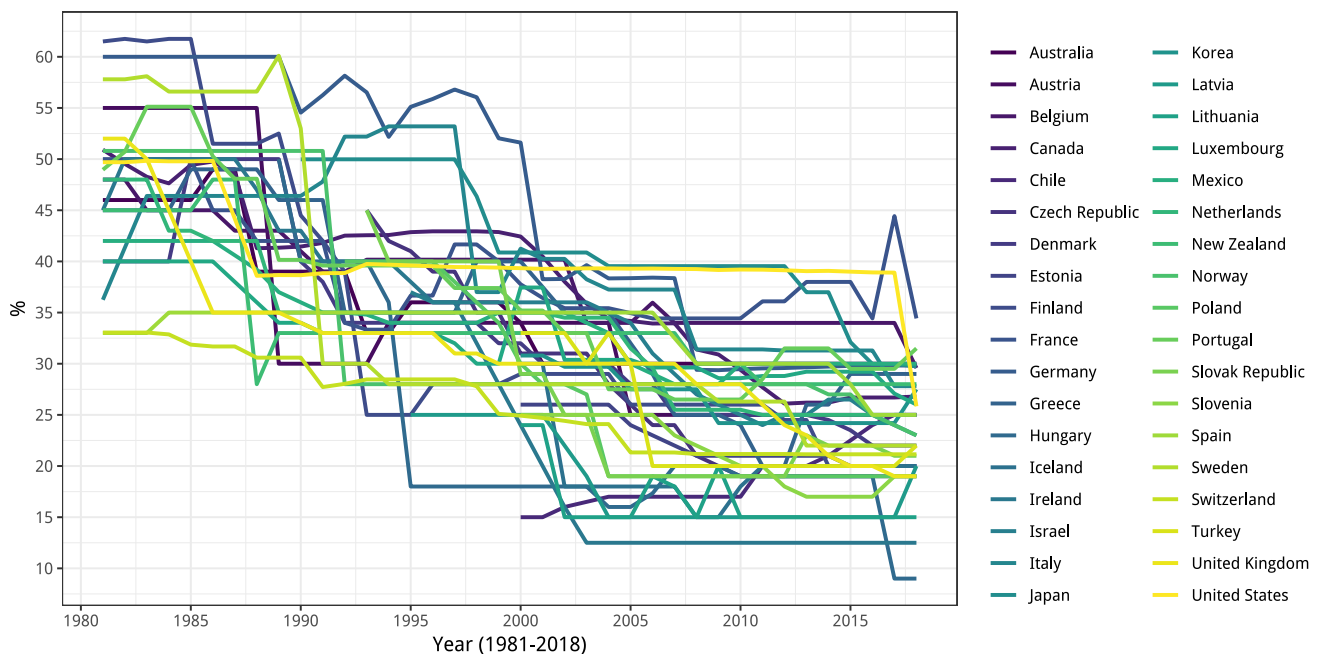
Note: Responses to a 14 percentage-point tax cut from the Southern country. Northern country retaliates with $\text{Corr}(\varepsilon_{N,t}^T, \varepsilon_{S,t}^T) = 0.8$.

Figure C.2: Welfare Responses to a Southern Tax Cut - with Retaliation from the Northern Country



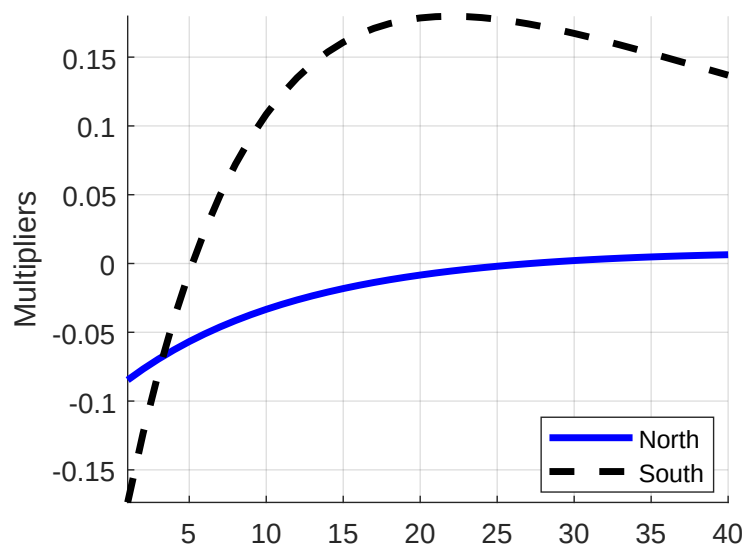
Note: Responses to a 14 percentage-point tax cut from the Southern country. Northern country retaliates with $\text{Corr}(\varepsilon_{N,t}^T, \varepsilon_{S,t}^T) = 0.8$.

Figure C.3: Corporate Taxes in the OECD



Note: Statutory tax rates are obtained from the OECD's tax library database. <http://www.oecd.org/ctp/tax-database.htm> Accessed on December 2018.

Figure C.4: Tax Multipliers



Note: This figure plots the tax multiplier following an exogenous tax decrease of one percent of output under the benchmark model.

D Extension: Spillovers Effects to the Rest of the World

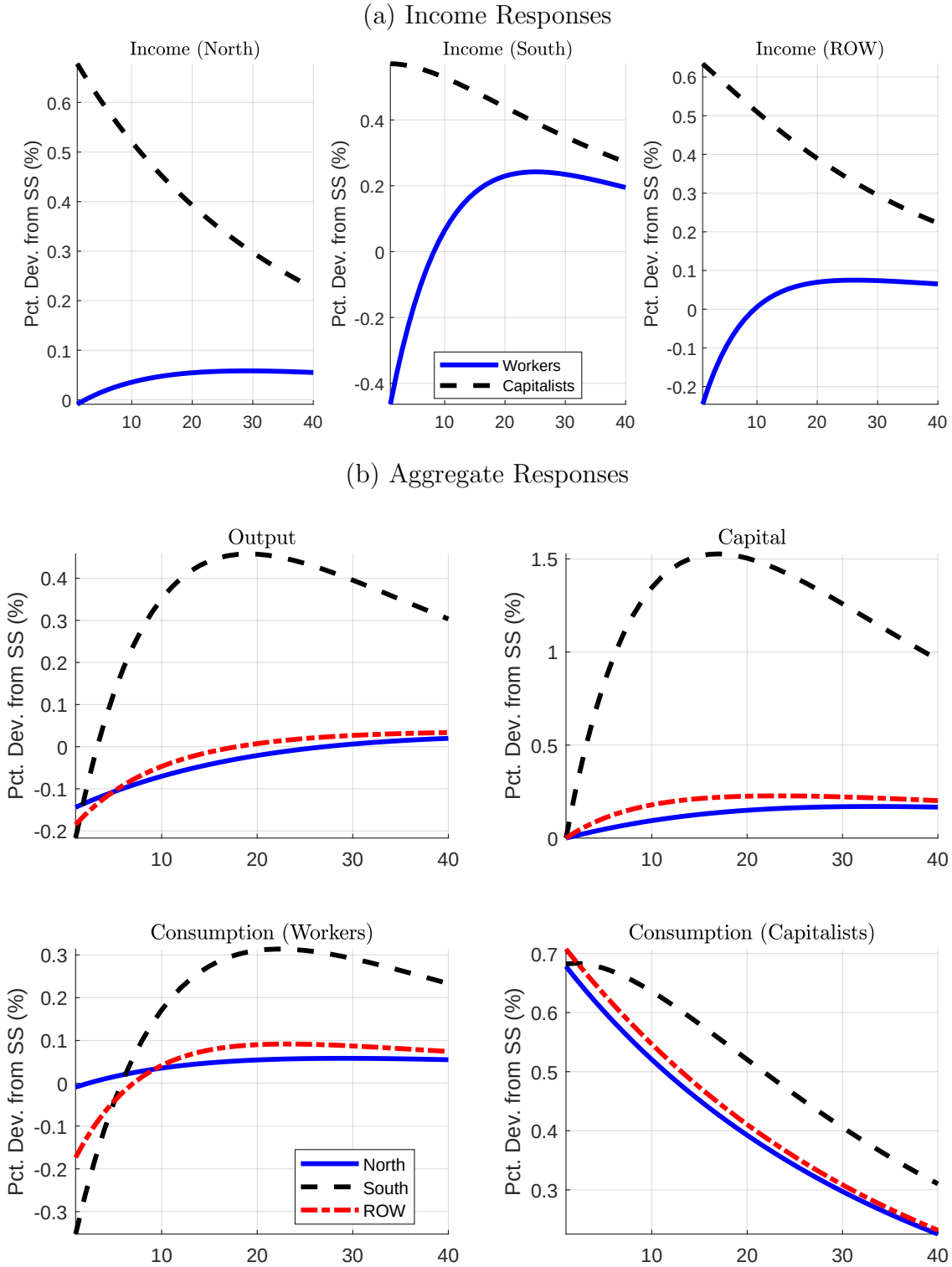
In this section, we explore the spillover effects of a CIT cut to the rest of the world by considering a three-country version of the benchmark model. We use an identical calibration strategy as in Section 3 for country one and country two (i.e., the North and the South). For the third country (i.e., the rest of the world), we use the same parameters for the South, except the state-state level of corporate income tax, which is set to be 24% to reflect the world's average corporate income tax.

[Figure D.1 Here]

We next examine the extent to which income for the capitalists and the workers in the North, the South, and the Rest of the World (ROW), along with several key macroeconomic aggregates, respond to a CIT cut of 14 percentage points that is initiated by the South. This value is motivated by the CIT cut arising from the Tax Cuts and Jobs Act of 2017 in the United States. We plot these responses, as the percentage deviation from the corresponding steady states, in Figure D.1.

One key insight from Figure D.1 is the stark contrast between the positive responses of capitalists' income and the negative responses of workers' income in the North, the South, and the Rest of the World. This result is consistent with the income responses to an identical CIT cut under the two-country version of the model. Intuitively, this result is driven by the increase in capital investment that arises from the increase in the marginal product of capital. Such an increase fuels output growth and since capitalists from all locations are able to invest in the firm in the South (where the tax cut was initiated), these agents are well-positioned to take advantage of the cut. This mechanism dovetails with the income responses observed in Figure D.1.

Figure D.1: Responses to a CIT cut from the South: Three-country Variant



Note: Responses to a 14 percentage-point tax cut from the Southern country under the three-country scenario. ROW denotes the rest of the world (i.e., the third country).